Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Mechatronics Engineering

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Course Code	:	ME 2130
Experiment No	:	01
Name of Experiment	:	Introduction to Basic Solid Modeling Techniques for 3D Part Creation
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Experiment No: 01

Name of Experiment: Introduction to Basic Solid Modeling Techniques for 3D Part Creation

1.1 Objectives

1. To become familiar with the user interface and basic functionalities of SolidWorks.
2. To learn how to create and fully define 2D sketches using various sketching and dimensioning tools.
3. To understand and apply fundamental 3D modeling features, specifically the Extruded Boss/Base and Revolved Boss/Base tools.
4. To practice creating complete 3D solid parts from given 2D drawings by following a step-by-step modeling process.

1.2 Theory

SolidWorks, by Dassault Systèmes, is a parametric Computer-Aided Design (CAD) software widely used for creating 3D solid models. The foundation of any model is a "fully defined" 2D sketch, where geometry is precisely controlled by dimensions and geometric relations [1]. This parametric approach, starting from a robust sketch, is fundamental to creating modifiable and predictable parts [2].

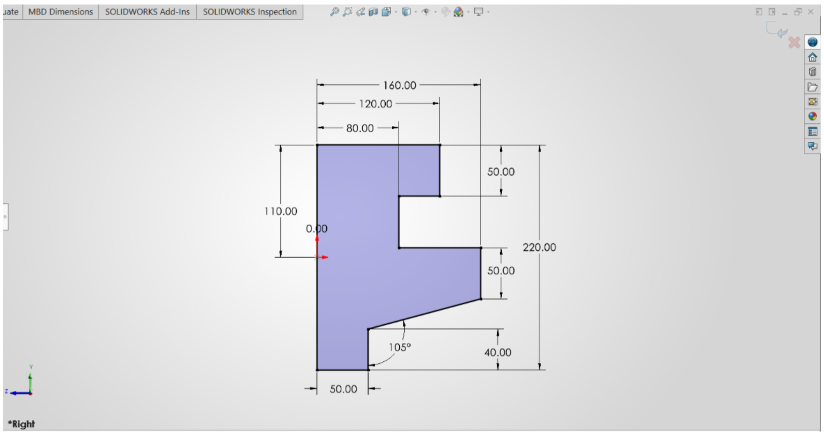


Figure 1.1: An example of a fully defined 2D sketch.

Once a 2D sketch is created, it can be transformed into a 3D part using various features. One of the most fundamental features is the Extruded Boss/Base. This tool adds depth to a selected sketch profile by projecting it linearly for a specified distance [3]. It is the primary method for creating prismatic parts and adding material based on a 2D shape [1].

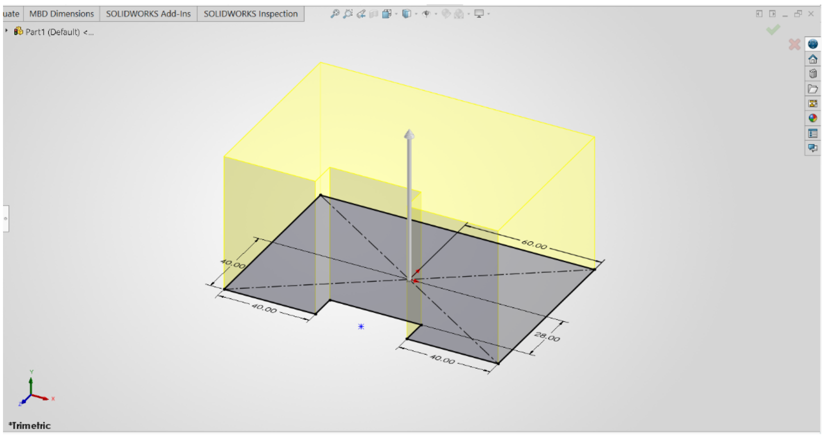


Figure 1.2: Illustration of the Extruded Boss/Base feature.

Another essential feature for creating cylindrical or axisymmetric parts is the Revolved Boss/Base. This tool creates a 3D model by revolving a 2D sketch around a specified centerline or axis [3]. It is highly efficient for modeling objects like wheels, shafts, and other parts with rotational symmetry. The ability to master both extrusion and revolution provides a powerful toolkit for modeling a vast array of mechanical components [2].

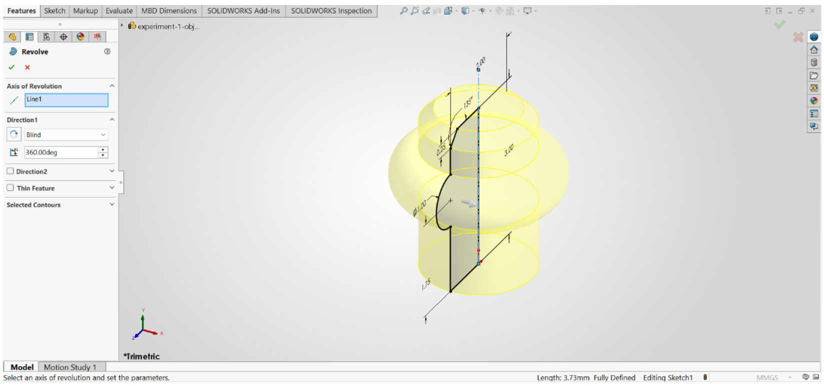


Figure 1.3: Illustration of the Revolved Boss/Base feature.

1.3 Apparatus Required

1. **Hardware:** A personal computer (PC) with hardware specifications sufficient for running SolidWorks smoothly.
2. **Software:** An installed and licensed version of the SolidWorks 3D CAD software by Dassault Systèmes.

1.4 Working

The creation of any 3D part in SolidWorks begins with a systematic workflow. This section outlines the initial steps required to set up the environment and create a basic 2D sketch, which forms the foundation for 3D features.

1.4.1 Unit System Selection

Upon launching SolidWorks and creating a new part file, the first crucial step is to define the unit system for the model. The dimensions of all sketches and features will be based on this selection. The unit system can be accessed and changed from the bottom-right corner of the status bar. For this lab, the MMGS (millimeter, gram, second) system is used for all models.

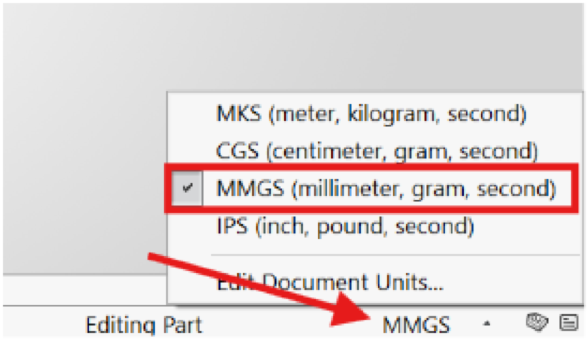


Figure 1.4: Selecting the MMGS unit system from the status bar.

1.4.2 Plane Selection

After setting the unit system, a base plane must be selected to host the initial 2D sketch. SolidWorks provides three default planes: Front, Top, and Right. The choice of plane is important as it determines the orientation of the part. To start a sketch, one must select a plane from the FeatureManager Design Tree or by clicking on it in the graphics area, and then select the Sketch command.

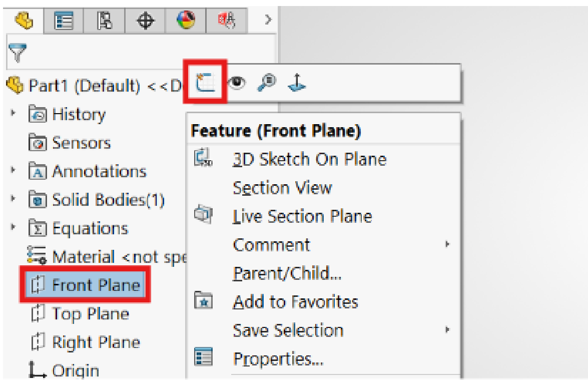
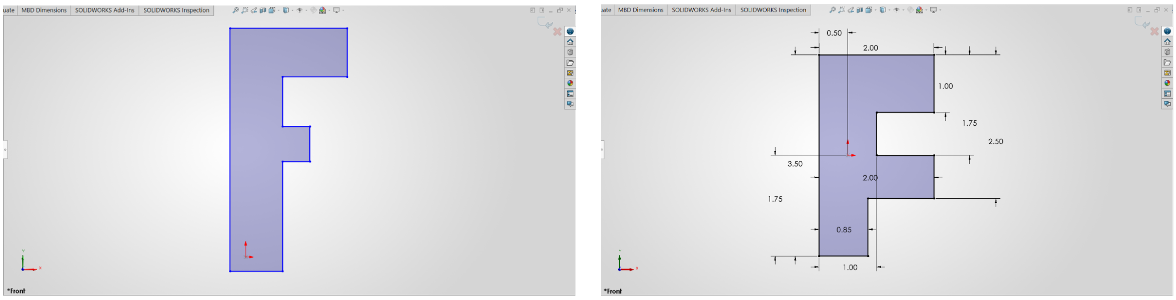


Figure 1.5: Selecting the Front Plane for the initial sketch.

1.4.3 Sketching and Dimensioning

With a plane selected and the sketch mode active, 2D geometry can be created using the tools in the Sketch CommandManager. Tools like Line, Circle, and Rectangle are used to draw the basic profile of the part.

After creating the geometry, it must be fully defined using dimensions and geometric relations. The Smart Dimension tool is used to add precise dimensional values to the sketch entities. Adding relations like Vertical, Horizontal, or Tangent helps to lock the geometry's position and orientation. A fully defined sketch is indicated by all lines turning black, and the status bar will display "Fully Defined." This ensures that the model is robust and will behave predictably when changes are made.



(a) Drawing lines and shapes for the sketch profile. **(b)** Using Smart Dimension to fully define the sketch.

Figure 1.6: Process of creating and defining a 2D sketch.

1.4.4 Extruded Boss/Base

Once a sketch is fully defined, the Extruded Boss/Base feature is used to convert the 2D profile into a 3D solid. This tool adds material by projecting the sketch perpendicular to the sketch plane for a specified depth. Various end conditions, such as *Blind*, *Through All*, and *Mid Plane*, offer control over the extrusion, making it a versatile tool for creating a wide range of prismatic parts.

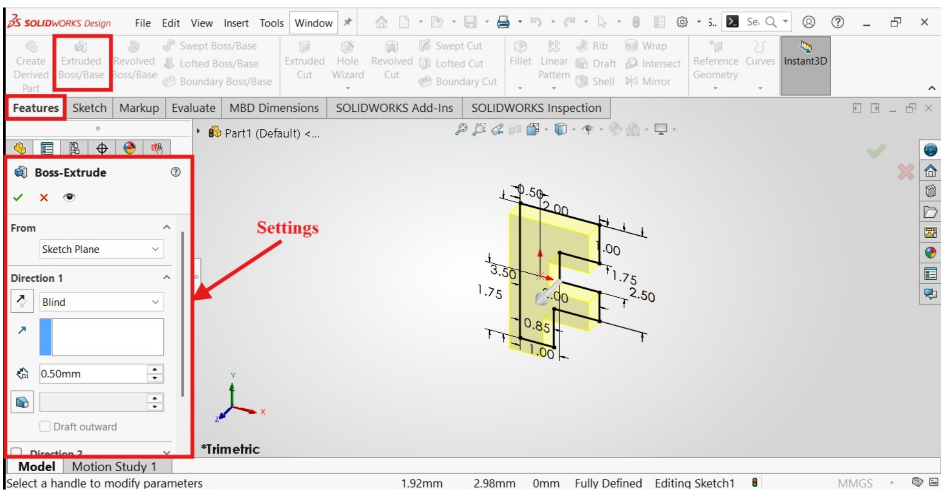


Figure 1.7: Applying the Extruded Boss/Base feature to a 2D sketch.

1.4.5 Revolved Boss/Base

For parts with cylindrical or axisymmetric geometry, the Revolved Boss/Base feature is used. This tool creates a 3D solid by revolving a 2D sketch around a selected centerline or axis. It is highly efficient for modeling objects such as wheels, shafts, and pulleys. The sketch profile defines the cross-section of the revolved part.

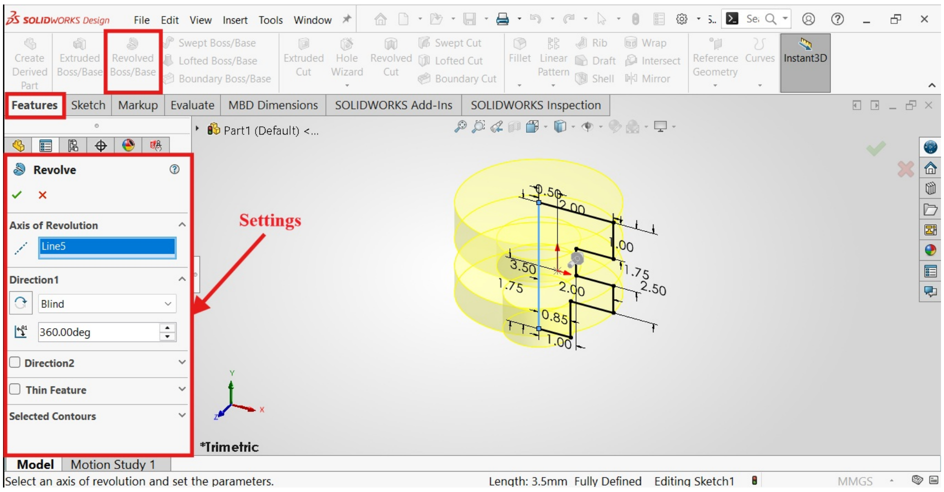


Figure 1.8: Using the Revolved Boss/Base feature with a sketch and centerline.

1.4.6 Fillet Feature

The Fillet feature is an applied feature used to create rounded internal or external faces along one or more edges. Fillets are essential for removing sharp corners, which can help reduce stress concentrations in mechanical parts and improve aesthetics. The feature allows for a constant or variable radius to be applied to the selected edges.

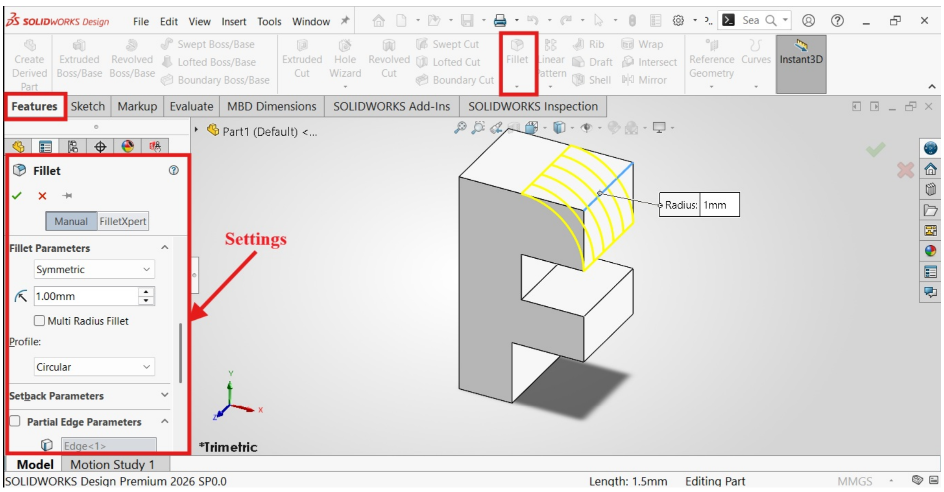
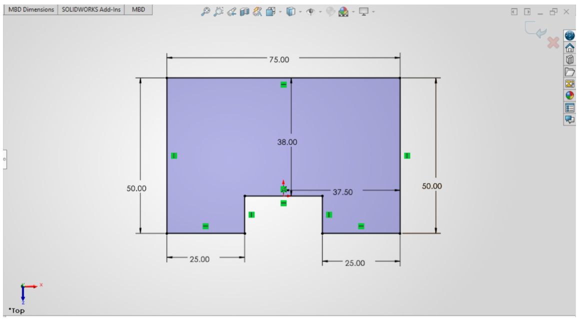


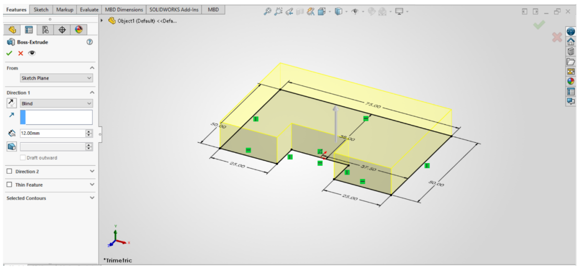
Figure 1.9: Applying a fillet to round sharp edges on a model.

1.5 Object 1

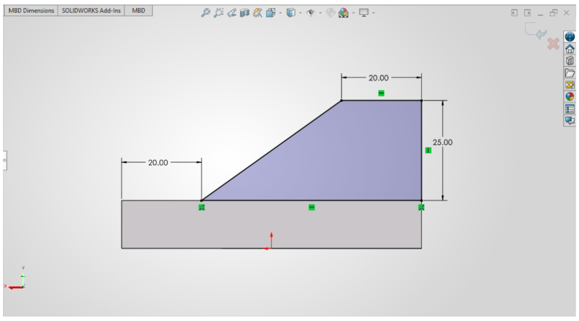
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



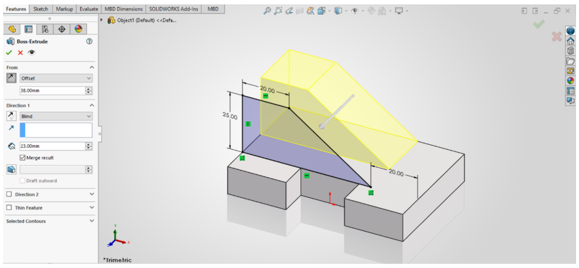
(a) 2D sketch of the base with dimensions



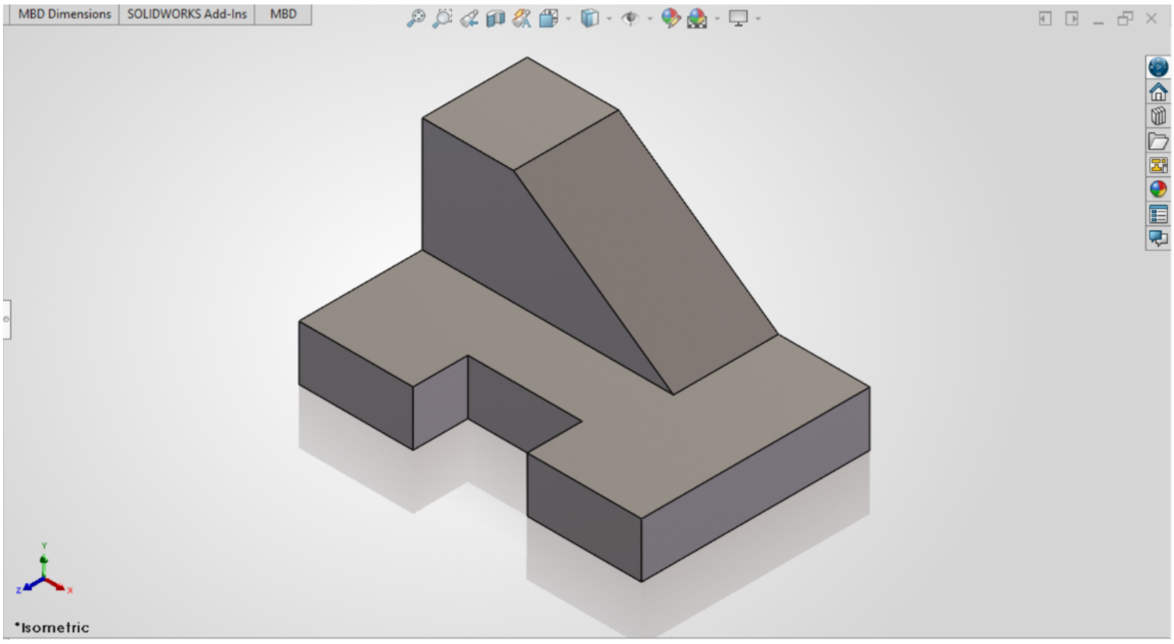
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion

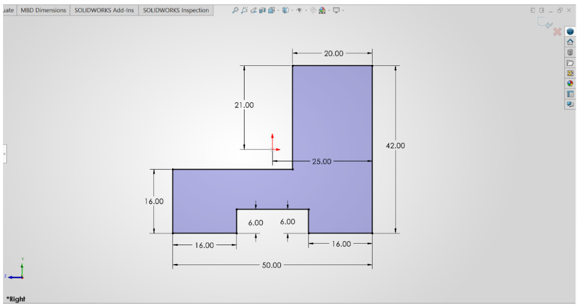


(e) Final result

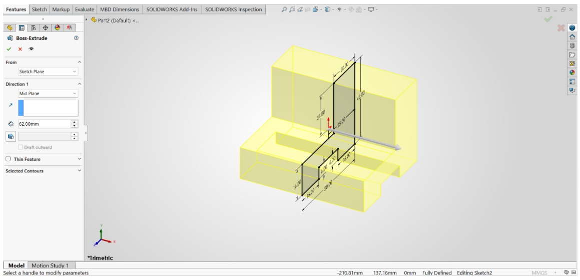
Figure 1.10: Snapshots while Designing Object 1 in Solidworks

1.6 Object 2

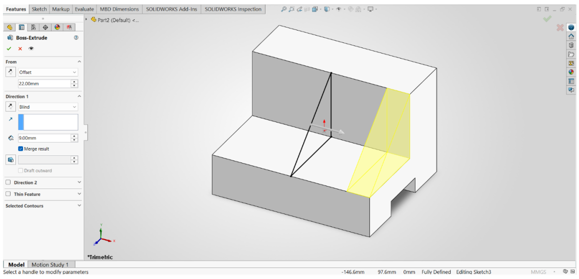
The construction of this object involved multiple features. The process began with a 2D sketch of the main body's profile, which was then extruded symmetrically using the "Mid Plane" end condition. Subsequently, a sketch for the top extension was created and extruded. The final step involved creating a sketch on the bottom face of the model and using the "Extruded Cut" feature to remove material and create the final shape.



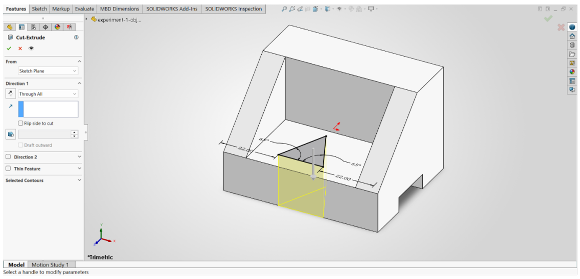
(a) 2D sketch of the main body with dimensions.



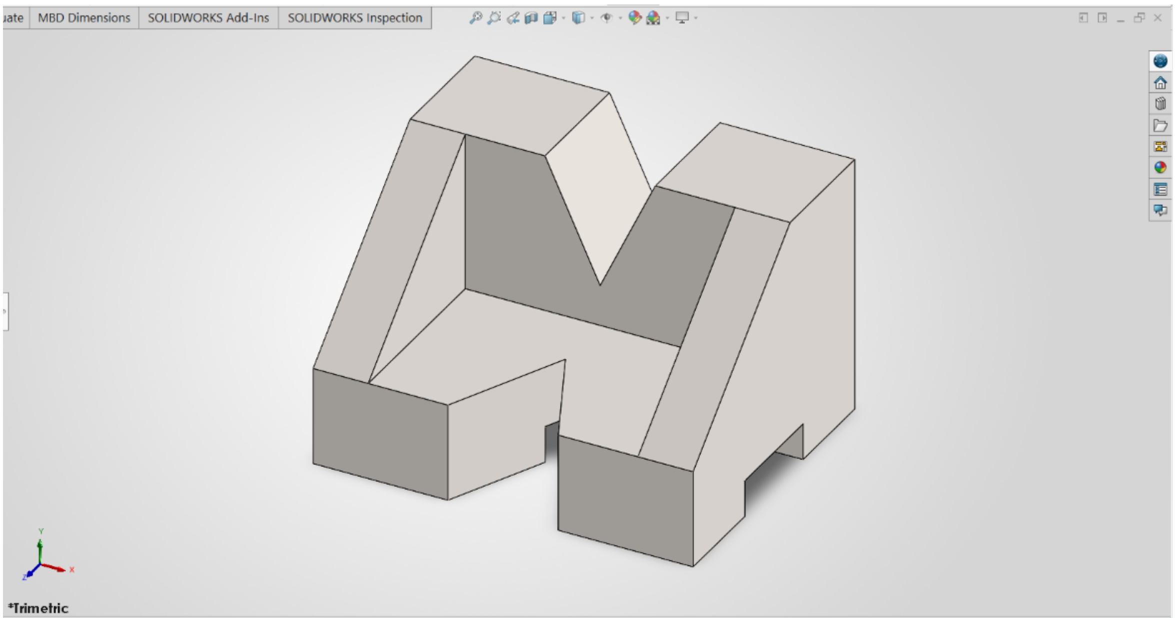
(b) Extruding the sketch from the mid-plane.



(c) Extruding the top extension.



(d) Applying the Extruded Cut feature.

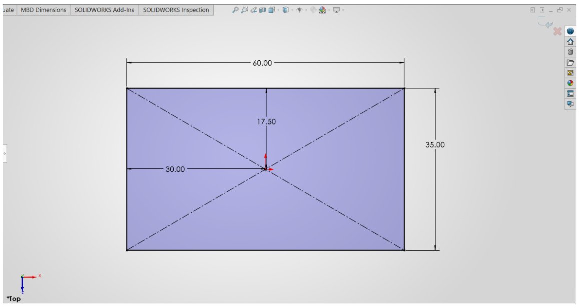


(e) Final Result.

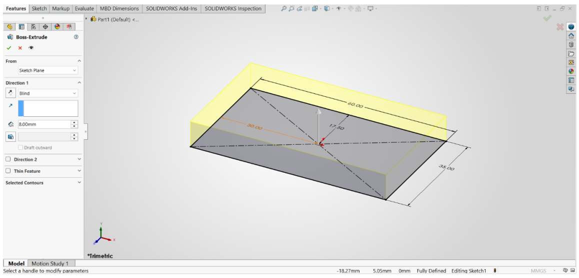
Figure 1.11: Snapshots while Designing Object 2 in Solidworks

1.7 Object 3

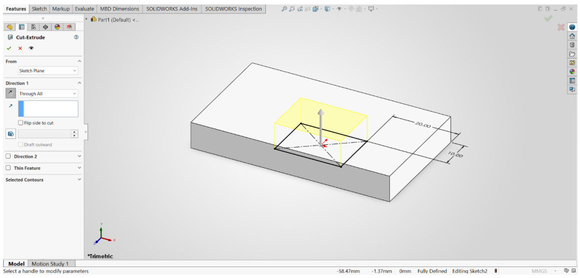
This object was created using a combination of additive and subtractive modeling techniques. The process began by sketching and extruding the main base. Following this, the ”Extruded Cut” feature was utilized to remove a rectangular section from the center. Finally, two smaller rectangular profiles were sketched on the top surface and extruded upwards to form the corner posts, completing the part’s geometry.



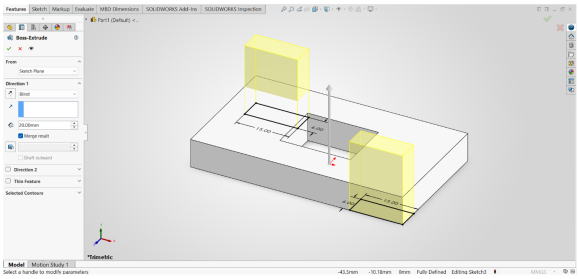
(a) 2D sketch of the base with dimensions.



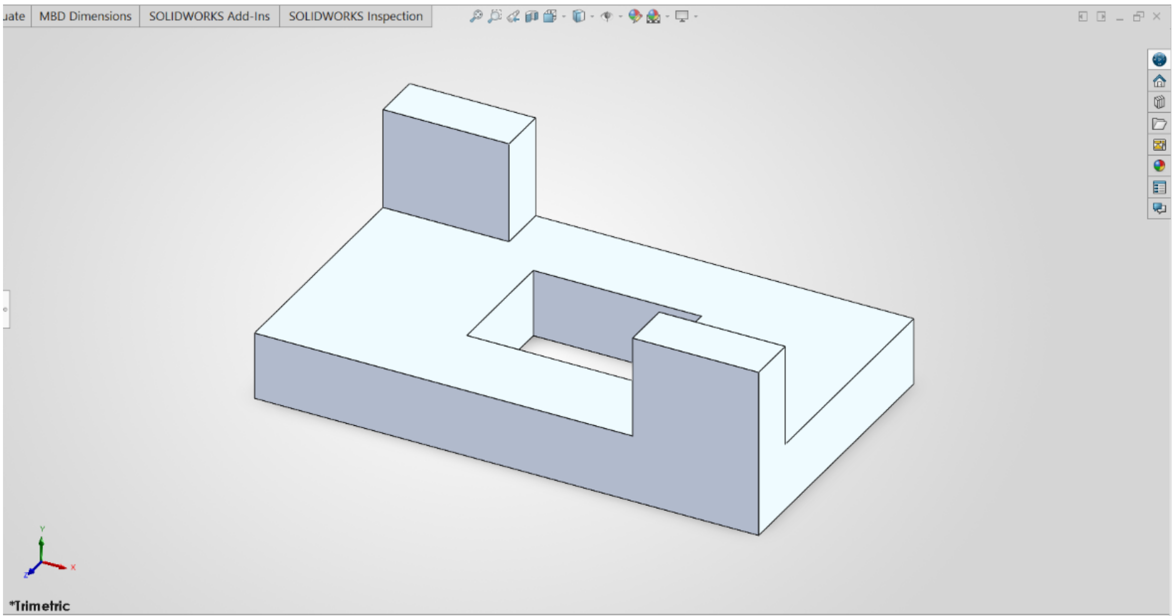
(b) Extruding the base sketch.



(c) Cutting a rectangle using the Extruded Cut feature.



(d) Extruding two rectangles from the corners.

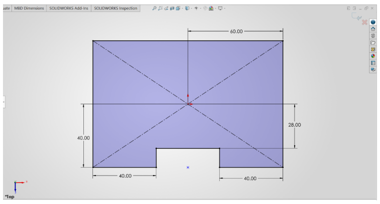


(e) Final Result.

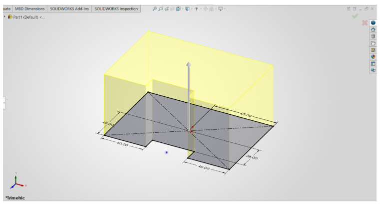
Figure 1.12: Snapshots while Designing Object 3 in Solidworks

1.8 Object 4

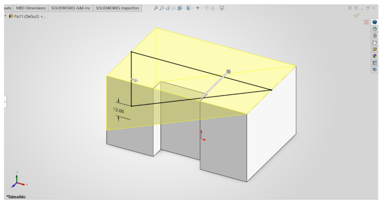
This part was constructed using a multi-step process involving both additive and subtractive methods. First, the main rectangular base was created by sketching its profile and applying an Extruded Boss/Base feature. Subsequently, a large angled cut was made using the Extruded Cut tool. A new sketch was then created on the top surface to extrude a central block upwards. The final step involved another Extruded Cut to remove a rectangular section from this newly created block, completing the model.



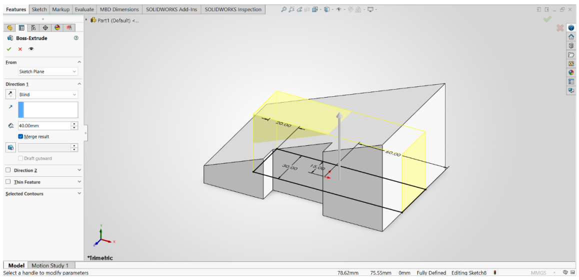
(a) 2D sketch of the base.



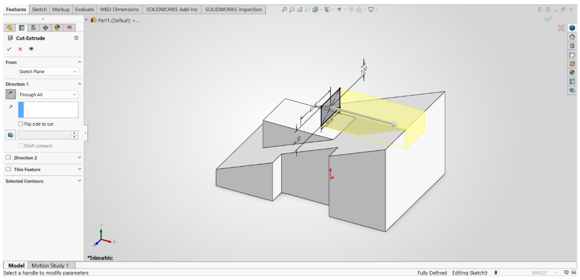
(b) Extruding the base sketch.



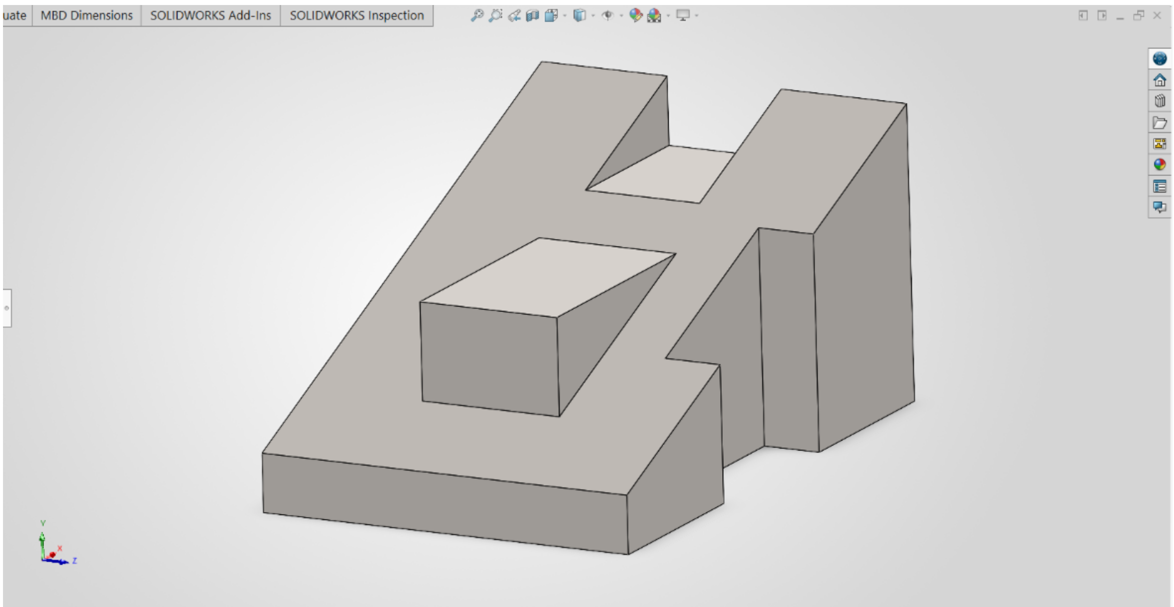
(c) Creating the angled cut.



(d) Extruding the central block.



(e) Creating the top cut-out.

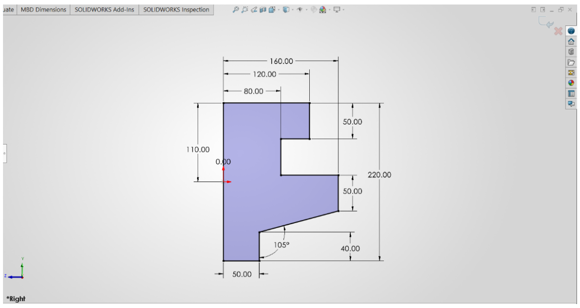


(f) Final result.

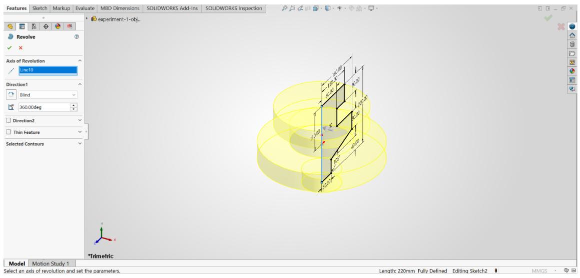
Figure 1.13: Snapshots while Designing Object 4 in Solidworks

1.9 Object 5

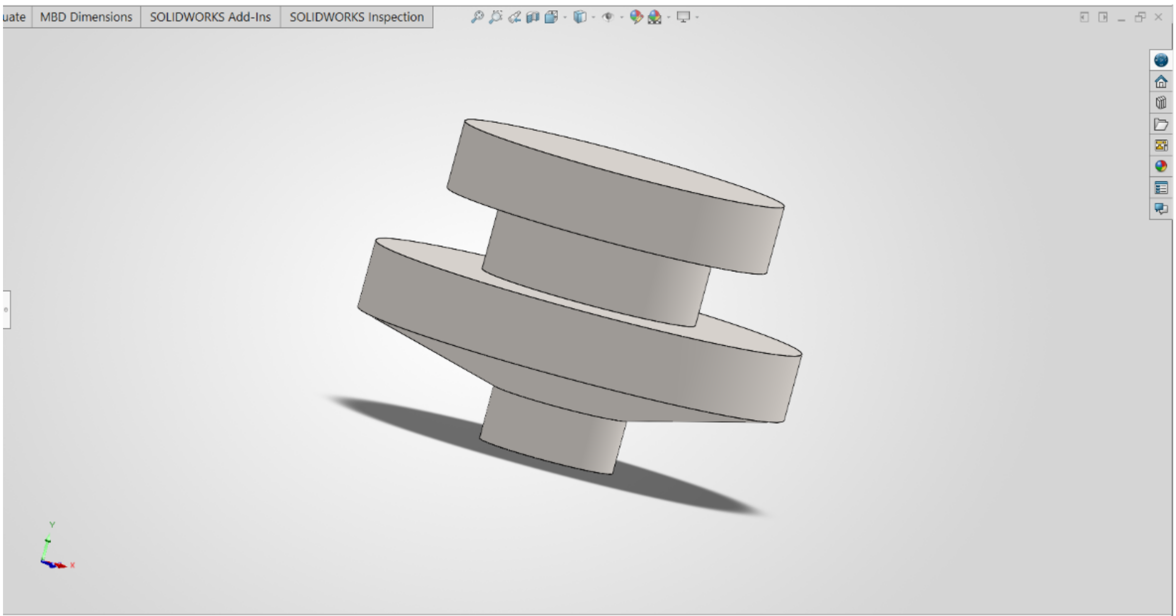
This part serves as a clear example of rotational modeling. A 2D cross-sectional profile was first sketched and dimensioned. The Revolved Boss/Base feature was then used to sweep this profile 360 degrees around a central axis, creating the complete, solid, axisymmetric geometry in a single step.



(a) 2D sketch with dimensions.



(b) Revolving the sketch.

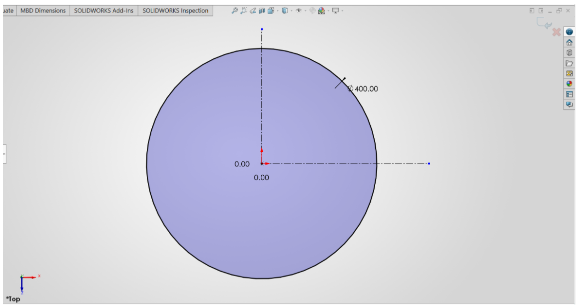


(c) Final Result.

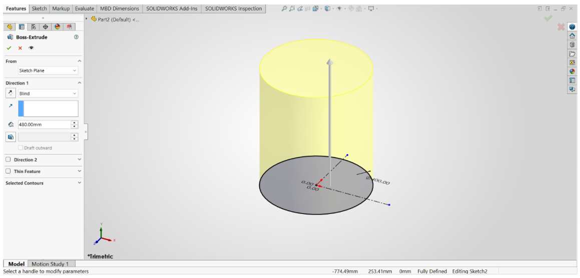
Figure 1.14: Snapshots while Designing Object 5 in Solidworks

1.10 Object 6

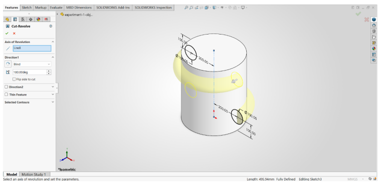
This object demonstrates the use of multiple feature types to build a complex part. The process began with creating and extruding the main base. Subsequently, a revolved feature was used to create circular geometry on two diagonal corners. To finalize the part, applied features were used: a Fillet was added to round a top edge, and a Chamfer was applied to create a beveled bottom edge, showcasing how different tools are combined in modeling.



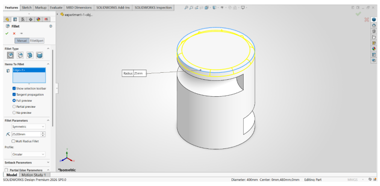
(a) 2D sketch of the base with dimensions.



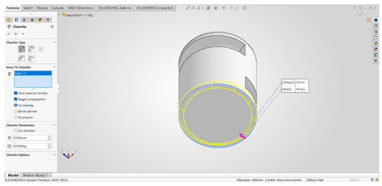
(b) Extruding the base sketch.



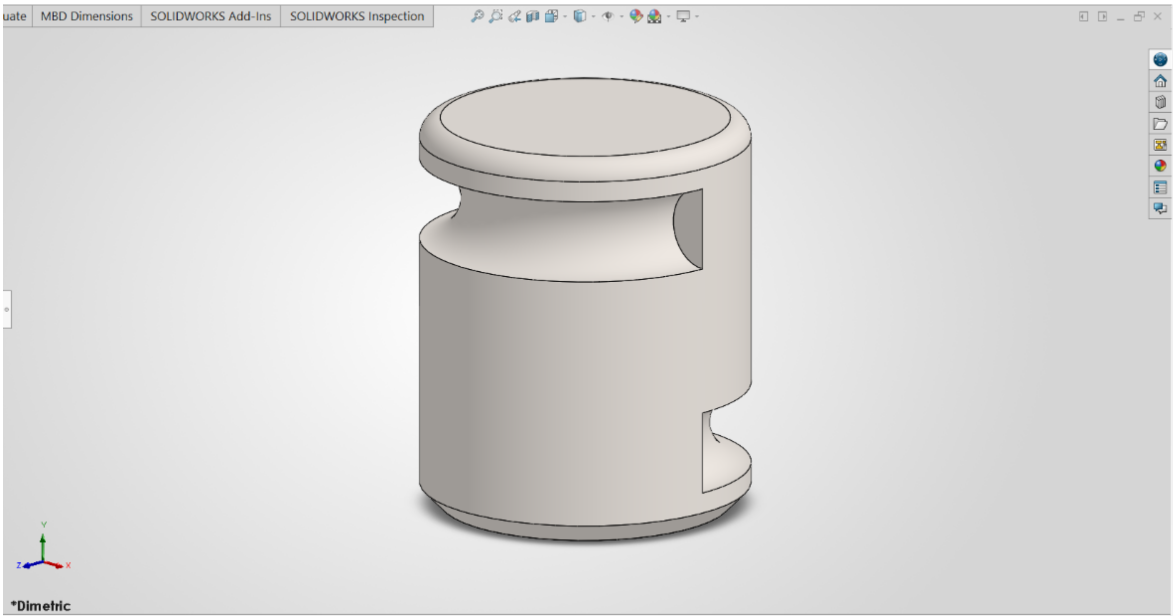
(c) Creating revolved features on diagonal corners.



(d) Applying a Fillet feature to the top edge.



(e) Applying a Chamfer feature to the bottom edge.

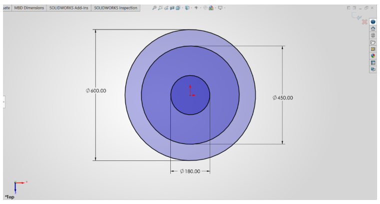


(f) Final Result.

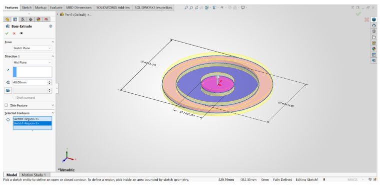
Figure 1.15: Snapshots while Designing Object 6 in Solidworks

1.11 Object 7

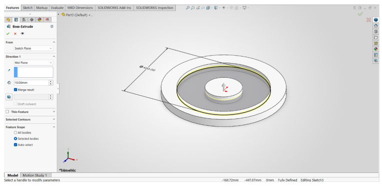
This object was constructed using a series of extrusion and cut features. The process began with a sketch of several concentric circles. The outer and middle rings were extruded to form the main body, followed by a separate extrusion for the inner circle. A chamfer was then applied to the edges. Finally, the Extruded Cut feature was used to create the holes around the part, completing the model.



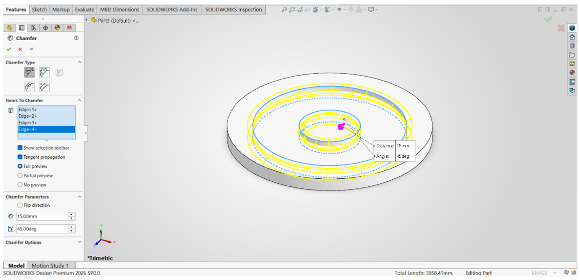
(a) 2D sketch of the base with dimensions.



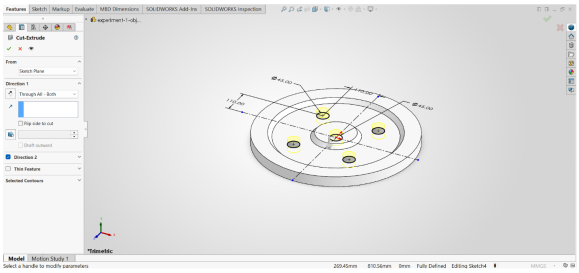
(b) Extruding the outer and middle circle.



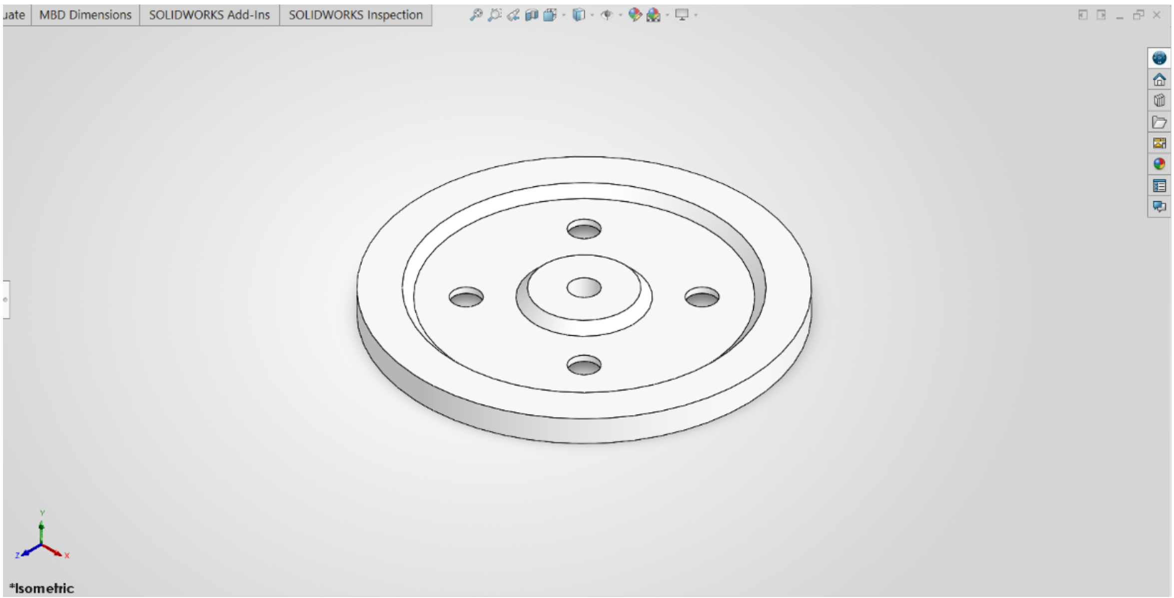
(c) Extruding the inner circle.



(d) Applying chamfer to the edges



(e) Cutting off the holes

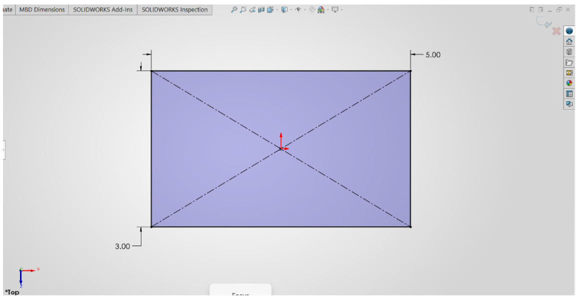


(f) Final result.

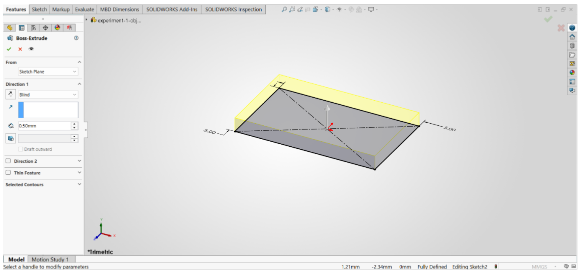
Figure 1.16: Snapshots while Designing Object 7 in Solidworks

1.12 Object 8

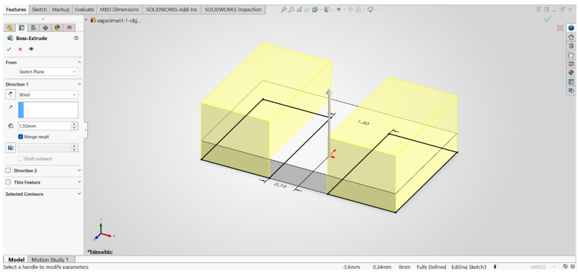
This part was created through a sequence of extrusion and cut operations. The process started with sketching and extruding the main base. Following that, two rectangular profiles were sketched and extruded from the sides. The final step involved using the Extruded Cut feature to create the circular holes, completing the model's geometry.



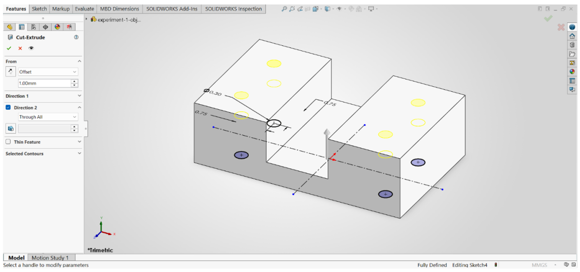
(a) 2D sketch of the base with dimensions.



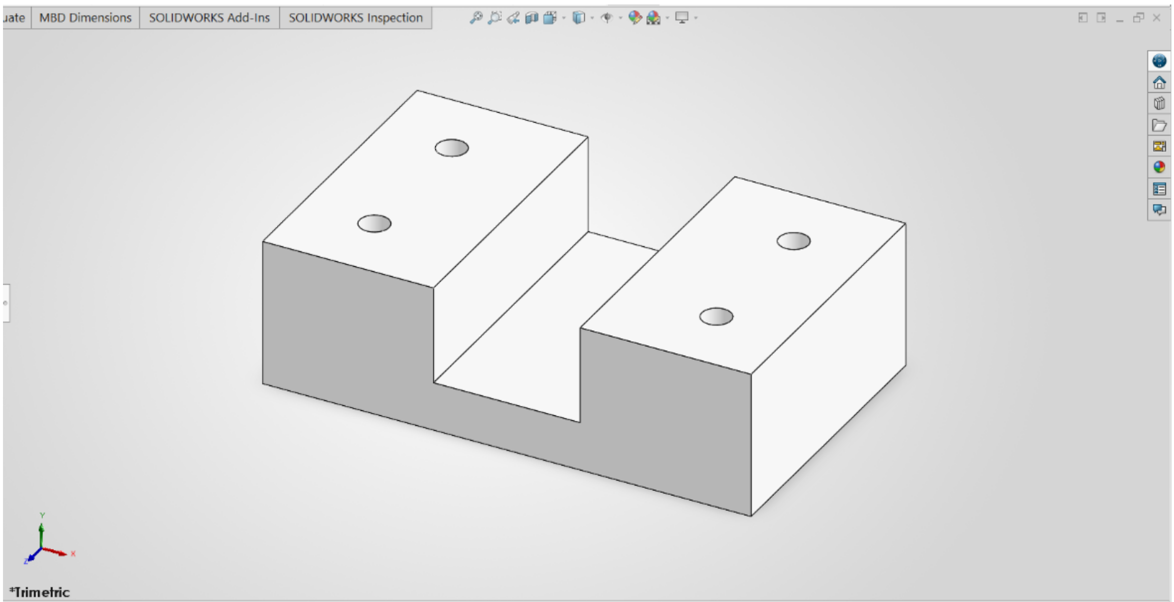
(b) Extruding the base.



(c) Extruding two rectangles from each side.



(d) Cutting off the holes.

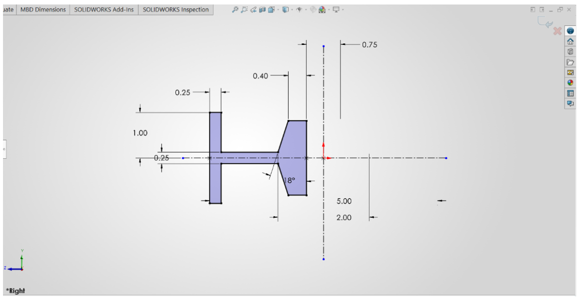


(e) Final result.

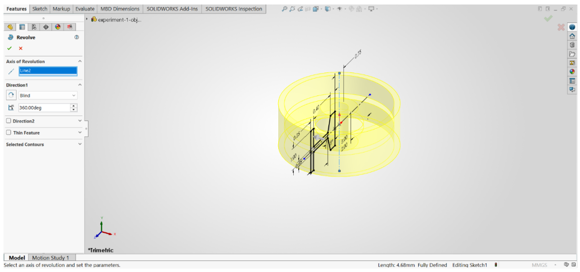
Figure 1.17: Snapshots while Designing Object 8 in Solidworks

1.13 Object 9

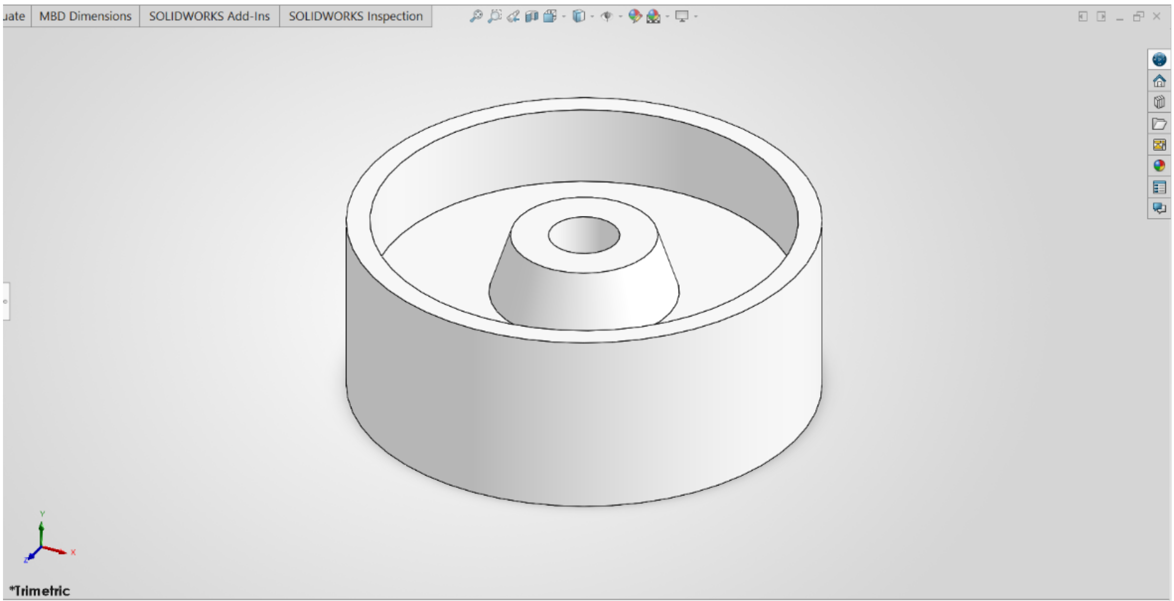
To create this component, the Revolved Boss/Base feature was employed. The key step was designing a precise 2D sketch that defined the part's cross-section. Once fully defined, this sketch was revolved around a vertical axis to produce the final three-dimensional shape, demonstrating an efficient method for creating parts with rotational symmetry.



(a) 2D sketch with dimensions.



(b) Revolving the sketch around the vertical axis.

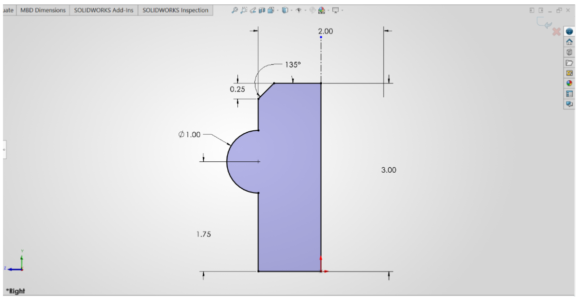


(c) Final Result.

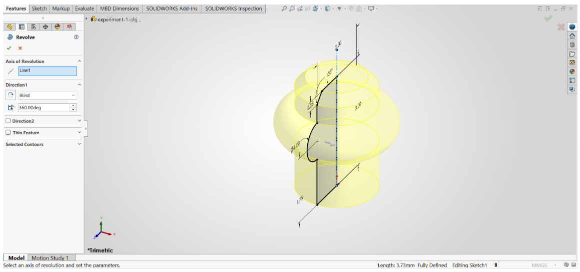
Figure 1.18: Snapshots while Designing Object 9 in Solidworks

1.14 Object 10

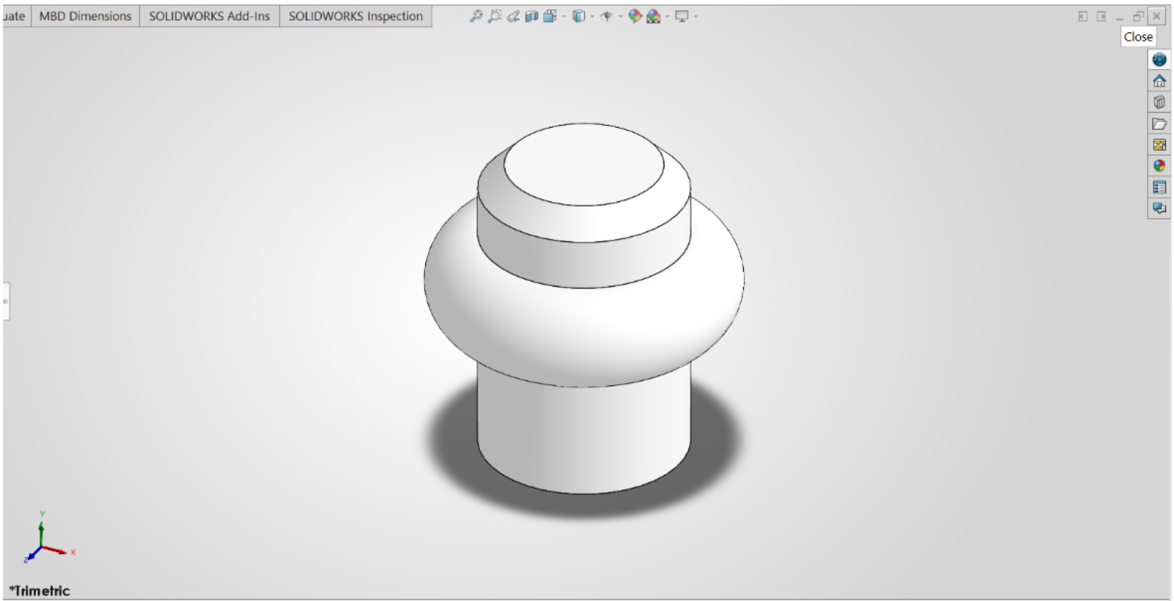
The geometry of this object was achieved using the Revolved Boss/Base tool. The process began with the creation of a detailed 2D sketch representing the part's profile. This sketch was then revolved around a central axis. This method is highly efficient for such axisymmetric parts, generating the complex form from a single profile sketch.



(a) 2D sketch with dimensions



(b) Revolving the sketch around the vertical axis



(c) Final Result

Figure 1.19: Snapshots while Designing Object 10 in Solidworks

1.15 Discussion

This lab provided practical experience in applying fundamental SolidWorks features to create a variety of parts. The objects demonstrated a clear distinction between modeling strategies: prismatic parts like Objects 1 and 2 were best suited for the Extruded Boss/Base feature, while axisymmetric parts like Objects 5, 9, and 10 were most efficiently created using the Revolved Boss/Base feature. This highlighted the importance of analyzing part geometry to select the optimal modeling approach. Furthermore, the process reinforced the necessity of creating fully defined 2D sketches as a robust foundation for all 3D features, which was critical for successfully modeling more complex parts involving multiple features like cuts, fillets, and chamfers.

1.16 Conclusion

In conclusion, this lab successfully met all its objectives by providing a comprehensive introduction to 3D part modeling in SolidWorks. Through the step-by-step creation of ten distinct objects, proficiency was gained in navigating the user interface, creating and fully defining 2D sketches, and applying fundamental features such as Extruded Boss/Base and Revolved Boss/Base. The experiment provided a solid foundation in parametric modeling principles and demonstrated the workflow for translating 2D drawings into complete 3D solid parts.

References

[1] M. Lombard, *SolidWorks Bible*, 2022nd. Hoboken, NJ: John Wiley & Sons, 2022.

[2] D. A. Madsen and D. P. Madsen, *Engineering Drawing and Design*, 6th. Clifton Park, NY: Cengage Learning, 2016.

[3] Dassault Systèmes. “3D CAD Software | SOLIDWORKS,” Accessed: Jul. 17, 2026. [Online]. Available: https://www.solidworks.com/product/3d-cad

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Mechatronics Engineering

Course Title	:	CAD Practice
Course Code	:	ME 2130
Experiment No	:	02
Name of Experiment	:	Application of Basic Solid Modeling Techniques for the Development of Three-Dimensional Parts
Date of Experiment	:	20 July 2026
Date of Submission	:	17 August 2026

Submitted By	**Submitted To**
Name : TASNIMUL HASAN Roll : 2408020 Series : 24	Dr. Dip Kumar Saha Assistant Professor Department of Mechatronics Engineering RUET Md. Faisal Rahman Badal Assistant Professor Department of Mechatronics Engineering RUET

Experiment No: 02

Name of Experiment: Application of Basic Solid Modeling Techniques for the Development of 3D Parts

2.1 Objectives

1. To model the seven assigned parts in SolidWorks using parametric workflows.
2. To produce fully defined 2D sketches and convert them into 3D geometry using features such as Extruded Boss/Base, Revolved Boss/Base, Cut, Fillet, Chamfer, and Shell.
3. To apply feature organization: mates, feature order, mirrors, and patterns to create robust, modifiable parts.
4. To document modeling decisions, common failure modes, and corrective steps for future design work.
5. To compare modeling strategies and justify the choice of features for each part with reference to CAD best practices [1], [2].

2.2 Theory

This lab continues the exploration of parametric solid modeling principles using SolidWorks [1], [3]. Central to parametric modeling is the notion of a fully defined sketch: a 2D profile constrained by dimensions and geometric relations so that subsequent features behave predictably [2].

Common features used in this experiment include:

- Extruded Boss/Base: creates prismatic geometry by giving a 2D profile a linear thickness. - Revolved Boss/Base: generates axisymmetric parts by revolving a profile about a centerline. - Cut operations: remove material from bodies using sketch-based profiles or intersecting geometry. - Fillets and Chamfers: apply local geometry changes to control stress concentrations and assembly fit. - Shell, Mirror, and Pattern features: promote reuse and maintainability of features across a part.

Selecting the correct combination of these tools reduces rebuild time and produces more robust models. For example, thin-walled parts benefit from the 'Shell' feature, while repeating geometry is best handled with 'Pattern' or 'Mirror' features. Fully defining sketches and naming important sketches/features aids later edits and collaboration [1], [2].

Figures in the lab document illustrate representative sketches and feature outcomes from the seven parts modeled in this experiment. Each part's section explains the chosen workflow and demonstrates how the features above were combined to produce the final geometry.

A central advantage of parametric modeling is the capture of design intent: dimensions and constraints express how a part should change when parameters are modified, enabling rapid iteration and variant generation [1]. Good sketching practice therefore emphasizes the use of driving dimensions for functional sizes, and geometric relations (coincident, concentric, tangent, perpendicular, parallel) to lock the sketch topology. Over-constraining a sketch can be as problematic as under-constraining it; the goal is clarity of intent rather than maximal constraint density [2].

Reference geometry (planes, axes, and construction lines) plays an important role when creating complex features like revolved profiles or lofts. Establishing named planes and sketch origins early prevents ambiguous references and reduces rebuild failures when the model history changes. Where repeated variations are required, 'Configurations' or design tables provide a controlled way to generate multiple part variants from a single master model, which is useful for family-of-parts exercises and assembly studies [3].

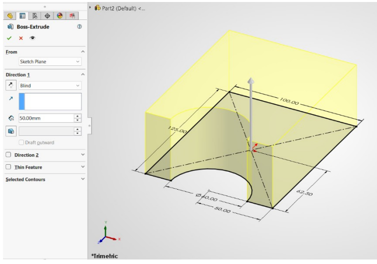
Finally, consideration of manufacturability and downstream processes should influence modeling choices. Simple features and regular faces ease CNC toolpaths and inspection; avoid unnecessarily intricate sketches when a combination of extrudes, cuts, and standard features (fillet/chamfer) will achieve the same form. Following these principles produces parts that are easier to edit, document, and prepare for production workflows [1], [2].

2.3 Apparatus Required

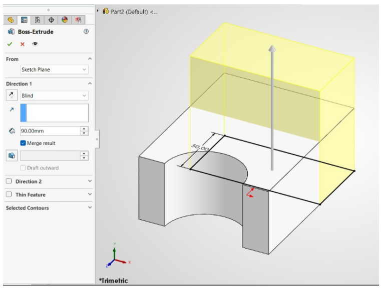
1. **Hardware:** A personal computer (PC) with hardware specifications sufficient for running SolidWorks smoothly.
2. **Software:** An installed and licensed version of the SolidWorks 3D CAD software by Dassault Systèmes.

2.4 Object 1

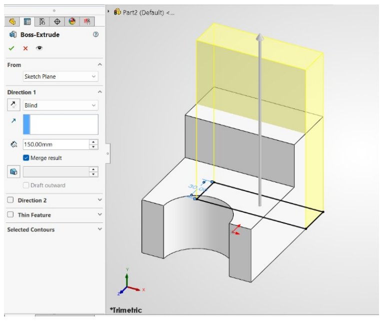
The base and arch profile were sketched and extruded to form the stepped block, after which an angled gusset rib was added with a second boss-extrude. A fillet was applied to the top edge of the arch, and a cut-extrude removed material through the arch to form the through-hole.



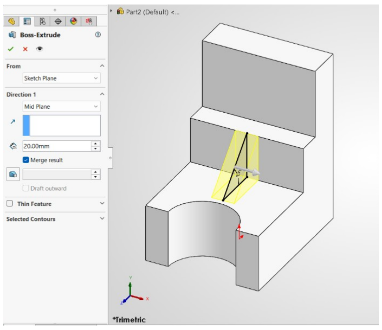
(a) Boss-extrude preview of the base/arch profile



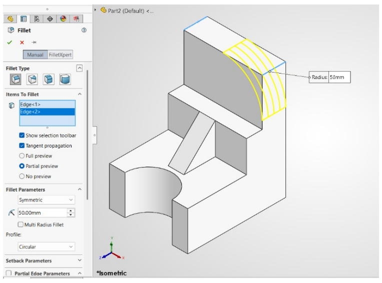
(b) Extruding the base block



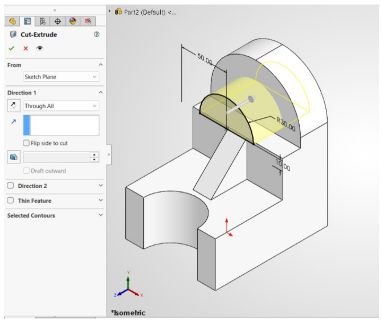
(c) Boss-extrude dialog for the upper block



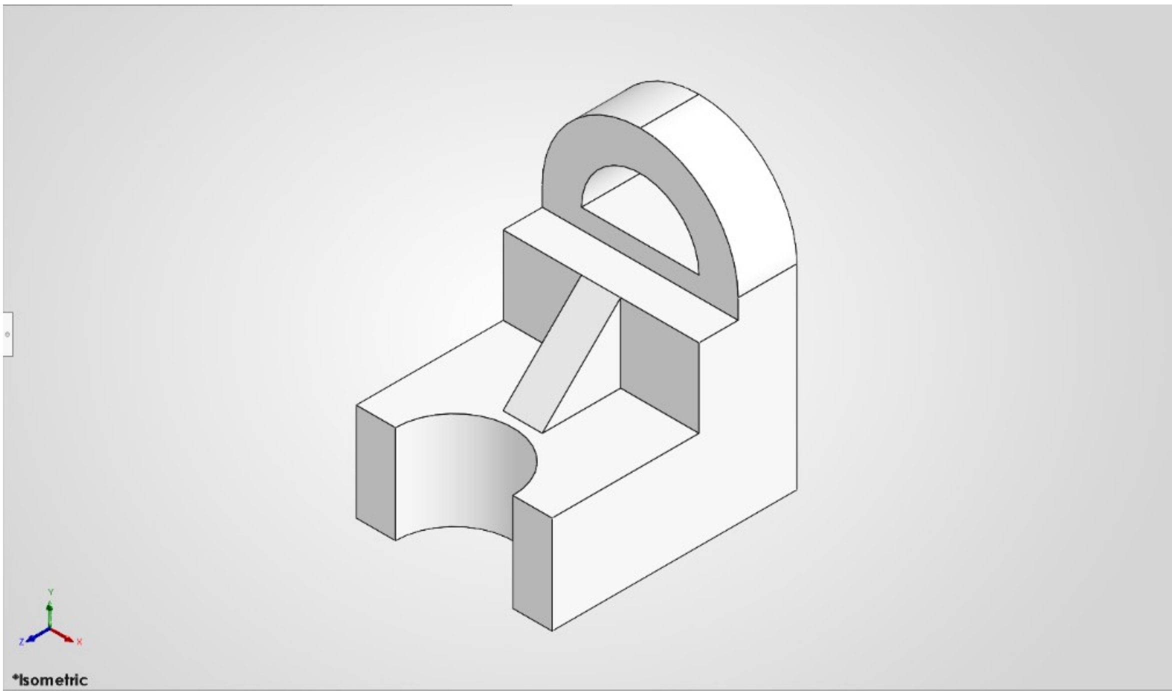
(d) Extruding the angled support rib



(e) Applying a fillet to the arch edge



(f) Cut-extrude forming the through-hole

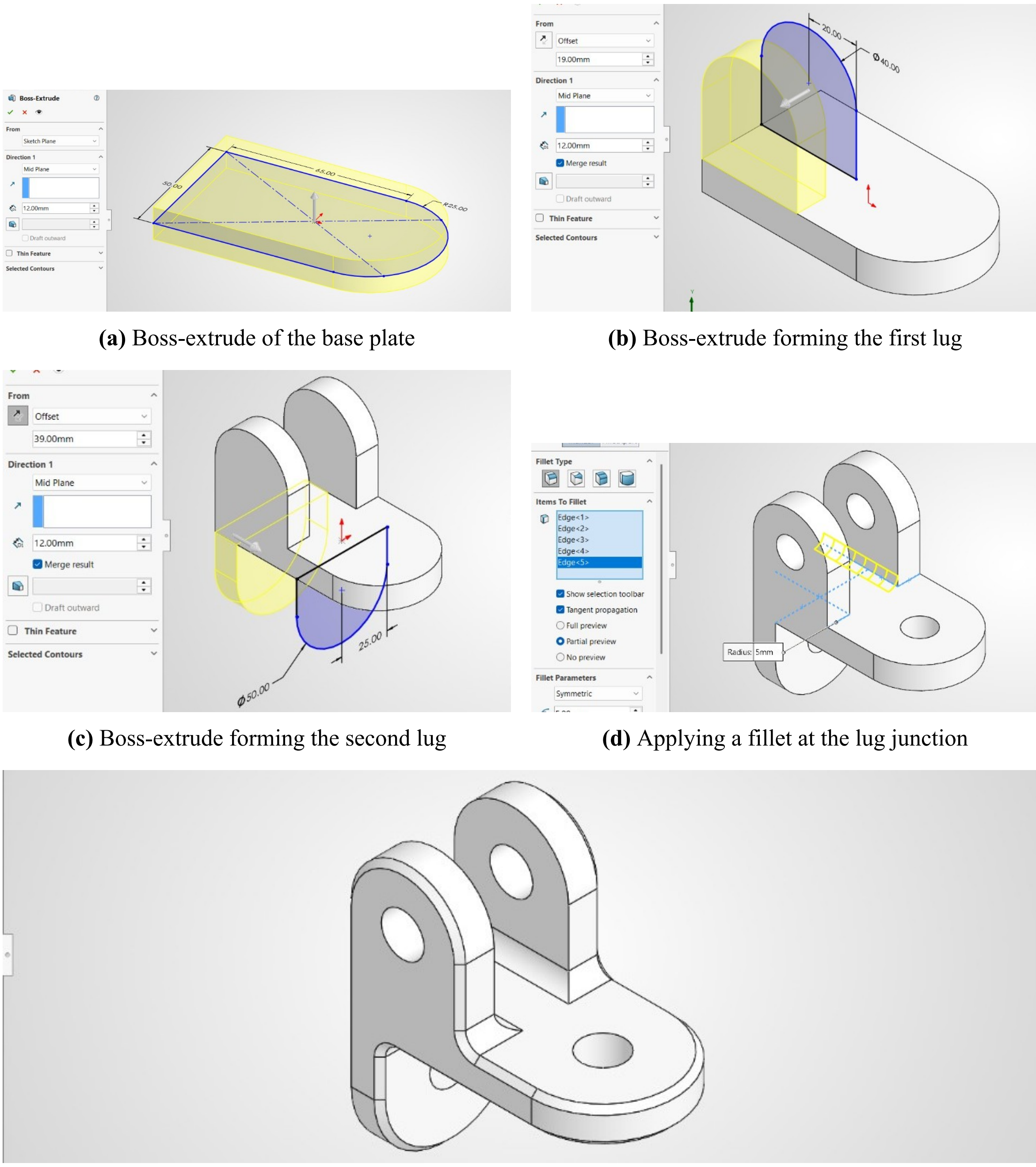


(g) Final result

Figure 2.1: Snapshots while Designing Object 1 in Solidworks

2.5 Object 2

The base plate was sketched and extruded, and a second boss-extrude formed the first rounded lug. A third boss-extrude added the second, overlapping lug, and a fillet was applied to blend the edges where the lugs meet the base.



(a) Boss-extrude of the base plate

(b) Boss-extrude forming the first lug

(c) Boss-extrude forming the second lug

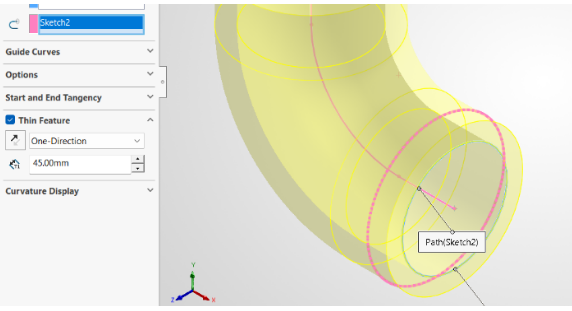
(d) Applying a fillet at the lug junction

(e) Final result

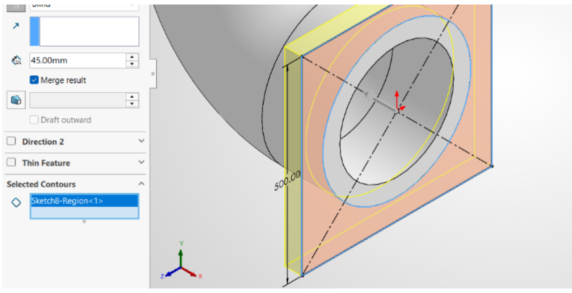
Figure 2.2: Snapshots while Designing Object 2 in Solidworks

2.6 Object 3

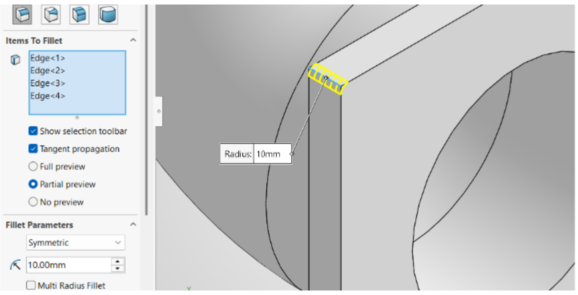
The elbow body was formed with a swept feature guided along a curved path, and a square flange was extruded onto one end. Fillets were applied to the flange edges, and a circular pattern replicated a cut feature to produce the four bolt holes.



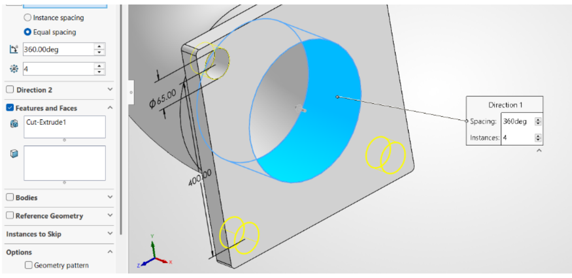
(a) Sweep dialog with guide curve/path for the elbow body



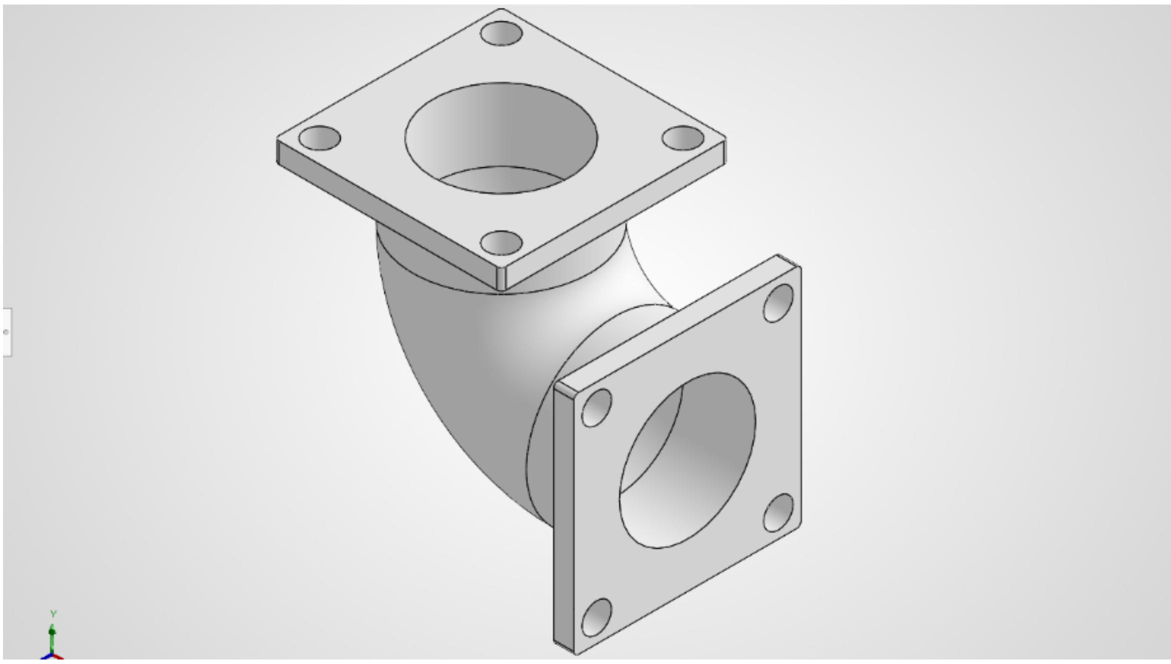
(b) Extruding the flange



(c) Applying fillets to the flange edges



(d) Circular pattern of the bolt holes

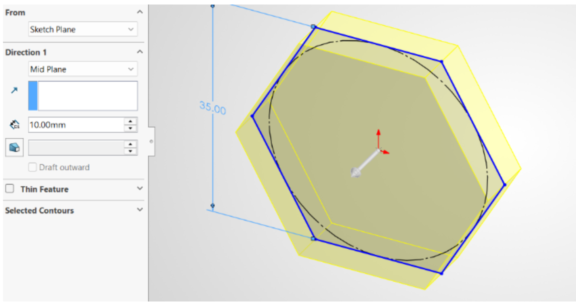


(e) Final result

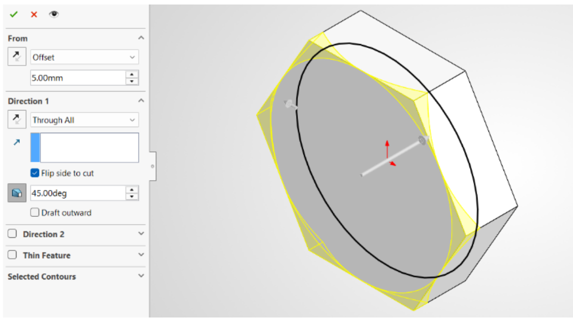
Figure 2.3: Snapshots while Designing Object 3 in Solidworks

2.7 Object 4

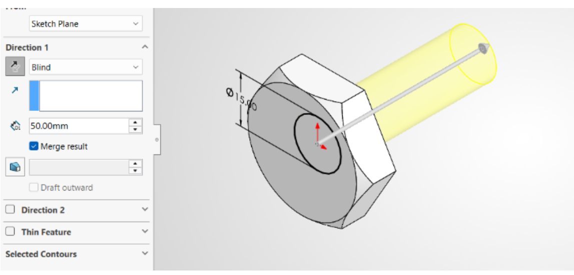
A hexagonal sketch was extruded to form the bolt head, and a cut-extrude chamfered its edge. A second sketch was extruded to form the cylindrical shank, and a thread feature was applied to generate the external threading.



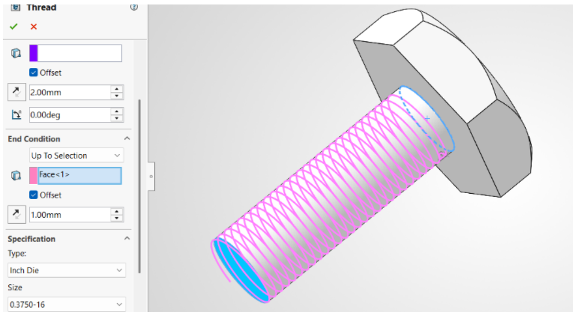
(a) Extruding the hex head



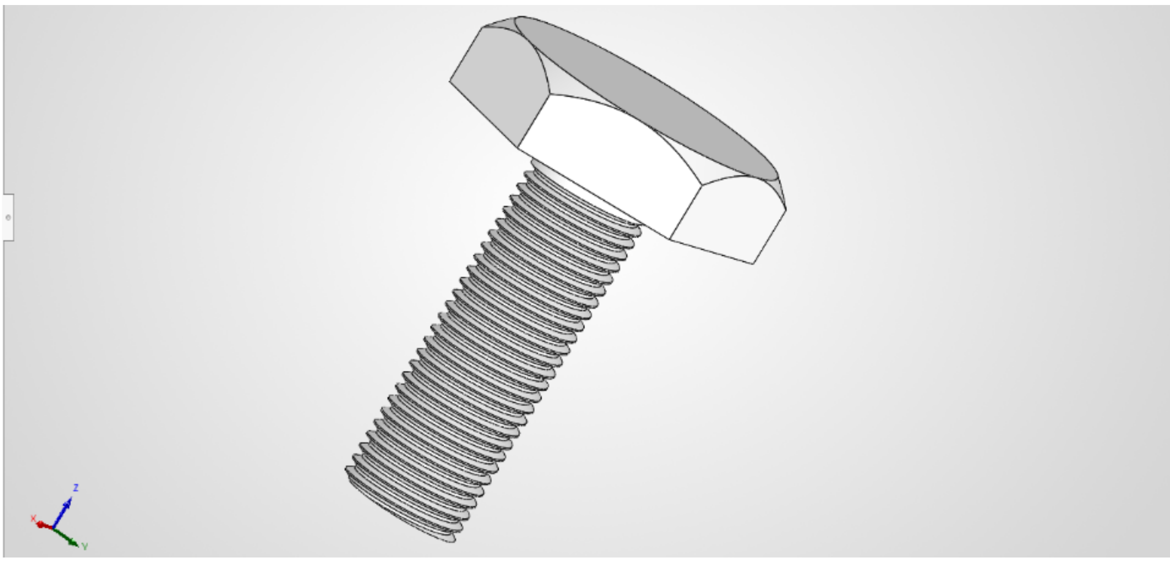
(b) Cut-extrude chamfering the head edge



(c) Extruding the cylindrical shank



(d) Applying the thread feature to the shank

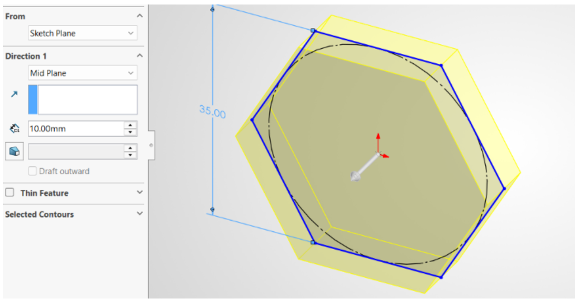


(e) Final result

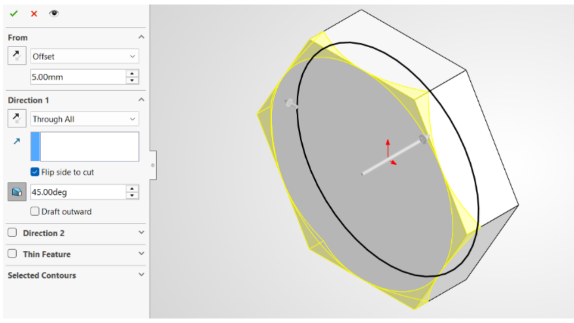
Figure 2.4: Snapshots while Designing Object 4 in Solidworks

2.8 Object 5

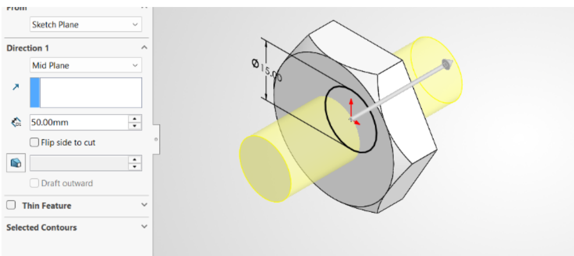
A hexagonal sketch was extruded to form the nut body, and a cut-extrude chamfered its edge. A further cut formed the central bore, and a tapped thread feature was applied inside the bore to generate the internal threading.



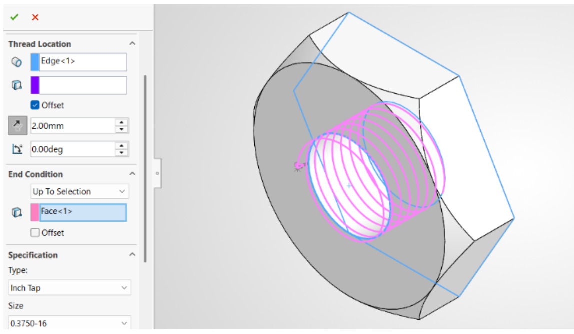
(a) Extruding the hex body



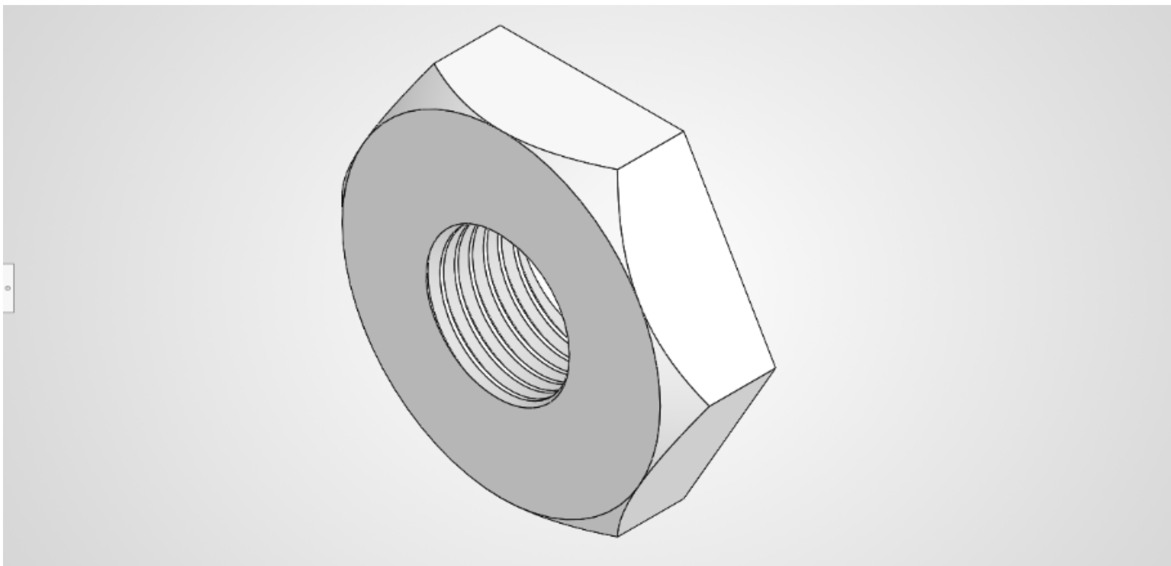
(b) Cut-extrude chamfering the body edge



(c) Cutting the central bore



(d) Applying the tapped thread to the bore

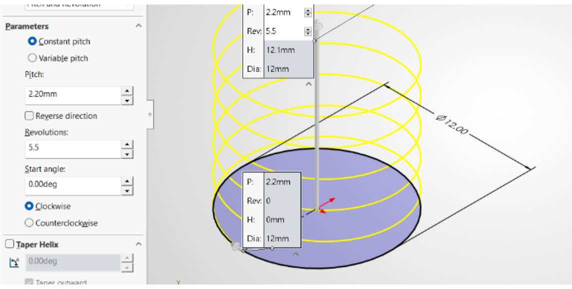


(e) Final result

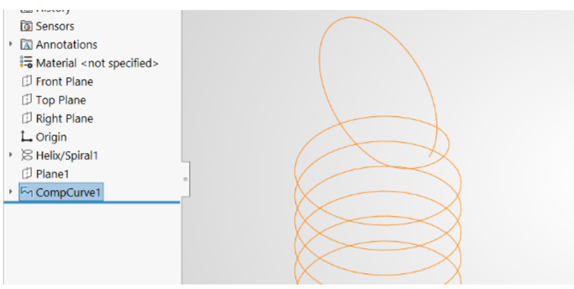
Figure 2.5: Snapshots while Designing Object 5 in Solidworks

2.9 Object 6

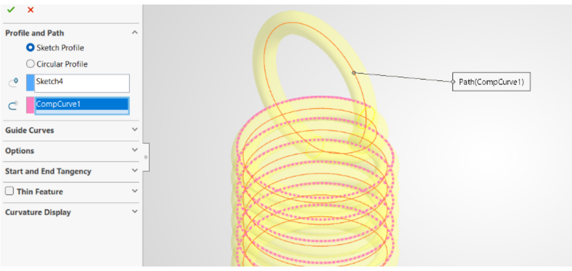
A helix/spiral was defined using pitch, height, and diameter parameters to generate the coil path, and a circular profile was swept along this path to form the first coil. The path and sweep were then rotated 180° and swept again to produce the second, interlocking coil.



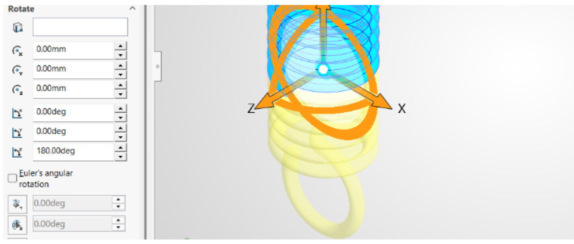
(a) Helix/spiral parameter dialog



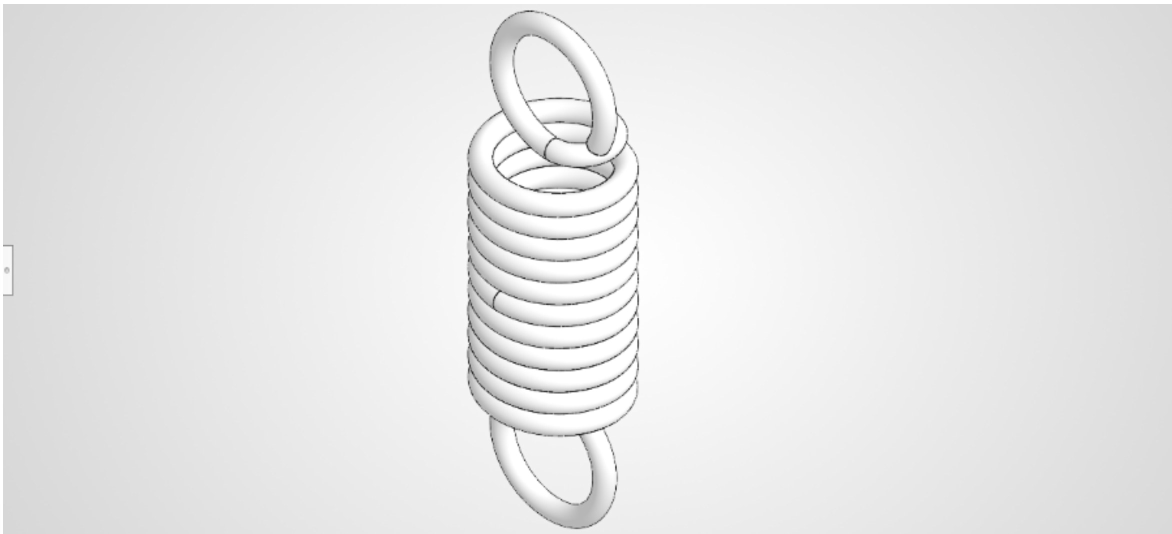
(b) Feature tree showing the generated helix curve



(c) Sweeping the circular profile along the path



(d) Rotating the path 180° for the second coil

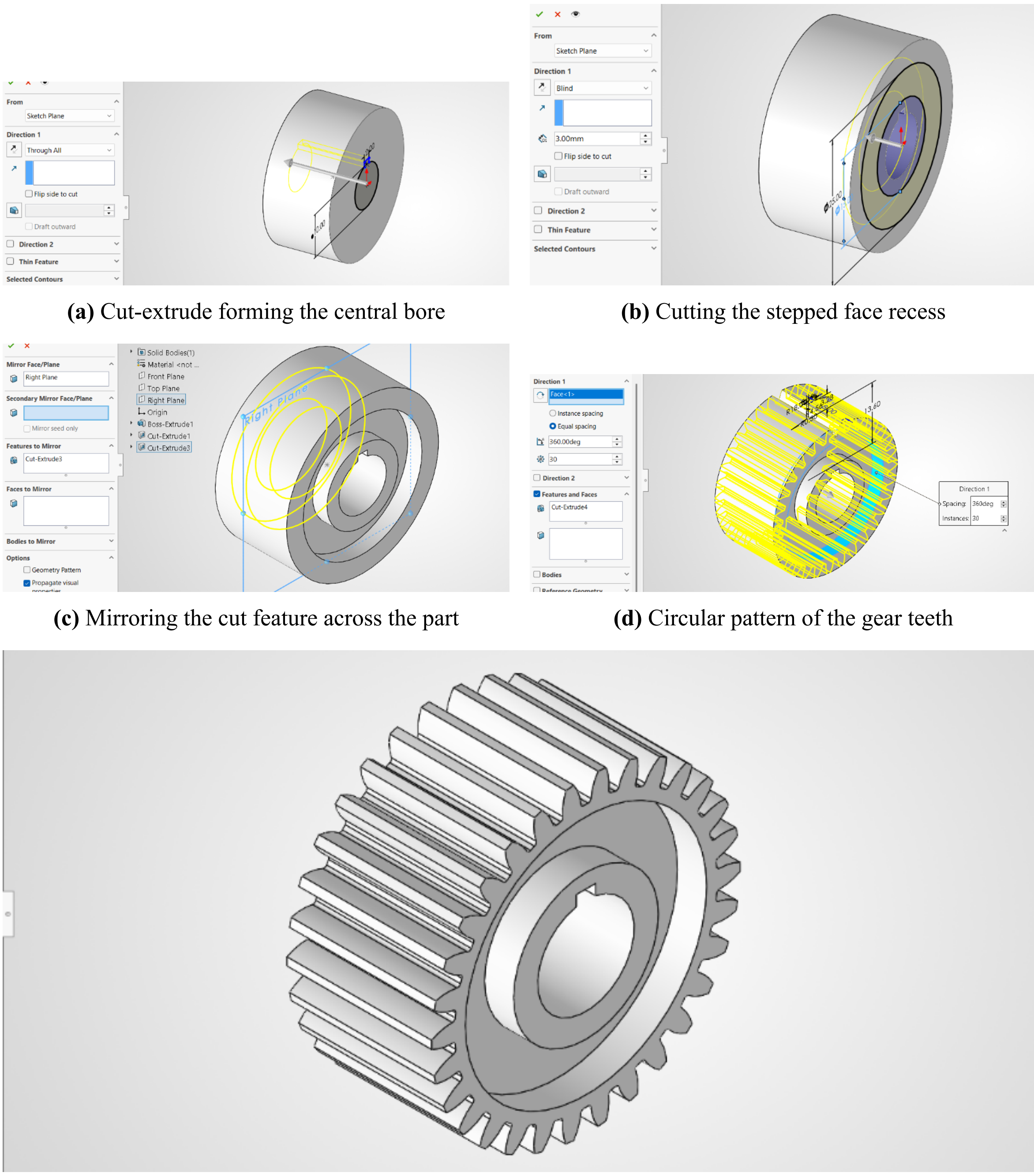


(e) Final result

Figure 2.6: Snapshots while Designing Object 6 in Solidworks

2.10 Object 7

A cut-extrude formed the central bore of the gear blank, and a second cut/extrude formed a stepped recess on the face. This cut was mirrored across the part, and a circular pattern (33 instances) replicated it around the full 360° to generate the gear teeth.



(a) Cut-extrude forming the central bore

(b) Cutting the stepped face recess

(c) Mirroring the cut feature across the part

(d) Circular pattern of the gear teeth

(e) Final result

Figure 2.7: Snapshots while Designing Object 7 in Solidworks

2.11 Discussion

The seven parts in this lab required a combination of prismatic and axisymmetric modeling strategies, and the experience reinforced the importance of selecting the correct feature sequence for each geometry. Parts with rectangular or stepped profiles were most effectively created using Extruded Boss/Base with subsequent cuts, fillets, or chamfers, while parts with rotational symmetry were best modeled with Revolved Boss/Base to maintain accuracy and efficiency. The process also showed that fully defined sketches are essential for stable geometry, because under-defined or poorly constrained sketches often lead to rebuild errors, ambiguity in dimensions, and difficulty when features need to be edited later. In addition, organizing the design tree, using reference geometry where necessary, and applying secondary operations after the main solid had been created improved the overall workflow and made the model easier to understand, revise, and document.

2.12 Conclusion

This lab successfully met its objectives by providing hands-on experience in parametric modeling with SolidWorks, where each part was developed through the careful creation of sketches, feature selection, and modification of geometry to match the required design. By completing the seven models, students strengthened their understanding of how to build robust parts from 2D profiles, apply essential operations such as extrusion, revolution, cuts, fillets, chamfers, and shells, and manage the feature history in a way that supports editing and future design changes. Overall, the exercise improved both technical modeling skills and the ability to think critically about design intent, geometry selection, and efficient modeling practices in CAD work.

References

[1] M. Lombard, *SolidWorks Bible*, 2022nd. Hoboken, NJ: John Wiley & Sons, 2022.

[2] D. A. Madsen and D. P. Madsen, *Engineering Drawing and Design*, 6th. Clifton Park, NY: Cengage Learning, 2016.

[3] Dassault Systèmes. “3D CAD Software | SOLIDWORKS,” Accessed: Jul. 17, 2026. [Online]. Available: https://www.solidworks.com/product/3d-cad

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Rajshahi University of Engineering and Technology

Department of Mechatronics Engineering

Course Title	:	CAD Practice
Course Code	:	ME 2130
Experiment No	:	03
Name of Experiment	:	Creation and Assembly of Mechanical Components Using SolidWorks
Date of Experiment	:	17 August 2026
Date of Submission	:	7 September 2026

Submitted By	**Submitted To**
Name : TASNIMUL HASAN Roll : 2408020 Series : 24	Dr. Dip Kumar Saha Assistant Professor Department of Mechatronics Engineering RUET Md. Faisal Rahman Badal Assistant Professor Department of Mechatronics Engineering RUET

Experiment No: 03

Name of Experiment: Creation and Assembly of Mechanical Components Using SolidWorks

3.1 Objectives

1. To design and model individual mechanical components using SolidWorks parametric modeling tools
2. To create dimensionally accurate parts with proper constraints and relationships
3. To assemble multiple components into functional assemblies and verify kinematic constraints
4. To apply engineering design principles including material selection, tolerance specification, and assembly considerations
5. To generate technical documentation including assembly drawings and bill of materials

3.2 Theory

Computer-aided design (CAD) software has become an indispensable tool in modern mechanical engineering for creating, analyzing, and documenting mechanical components and assemblies [1]. SolidWorks, a parametric 3D modeling platform, enables engineers to design components using feature-based modeling where geometric constraints and relationships control part dimensions. This parametric approach allows for rapid design iterations and ensures dimensional consistency across related components. The fundamental workflow involves sketching 2D profiles on selected planes, applying geometric and dimensional constraints to create the sketch, and then utilizing solid modeling features (such as extrude, revolve, sweep, and loft) to generate 3D part geometry. Each feature maintains a history-based relationship, allowing modifications to upstream features to automatically propagate through the design, thereby facilitating efficient design refinement and variant generation.

Assembly design in SolidWorks involves combining individual parts into a functional assembly through the application of kinematic constraints that define part relationships and restrict degrees of freedom [2]. Standard assembly constraints include coincident, parallel, perpendicular, and distance constraints, which together simulate the physical constraints and motion of real mechanical systems. Once parts are assembled with appropriate constraints, kinematic analysis can verify that components move as intended without interference or constraint conflicts. Technical documentation, including detail drawings with dimensions, tolerances, and notes, along with assembly drawings and bills of materials, constitutes the complete design deliverable [3]. This documentation serves as the communication bridge between design intent and manufacturing execution, ensuring that fabricated components meet functional and quality requirements.

3.3 Apparatus Required

1. **Hardware:** A personal computer (PC) with hardware specifications sufficient for running SolidWorks smoothly.
2. **Software:** An installed and licensed version of the SolidWorks 3D CAD software by Dassault Systèmes.

3.4 Object 1

Part 1

The primary structural component was modeled using basic sketching and extrusion features to establish the overall geometry. Dimensional constraints were applied to ensure precise dimensions, and fillets were added to reduce stress concentrations at sharp edges. This part serves as the foundational element upon which other components are mounted.

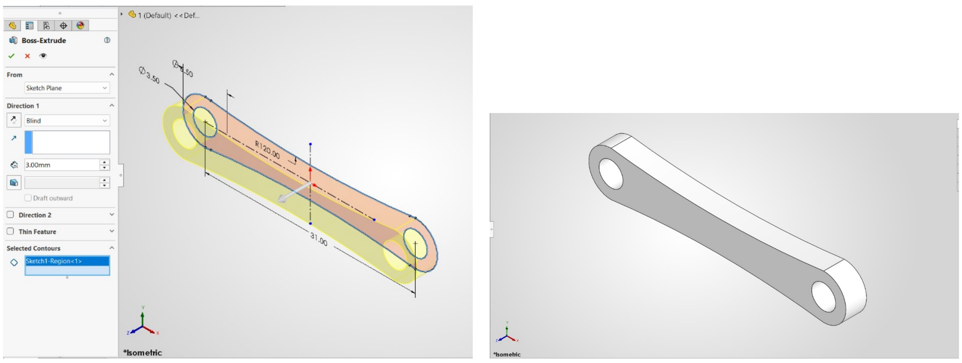

Figure 3.1: Snapshots while Designing Part 1 of Object 1 in Solidworks

Part 2

This component was designed with features that enable controlled motion and load transmission. Multiple circular profiles and stepped geometries were created using revolve and pocket features to achieve the required functionality. Tolerance specifications were incorporated to ensure proper fit and operation with mating components.

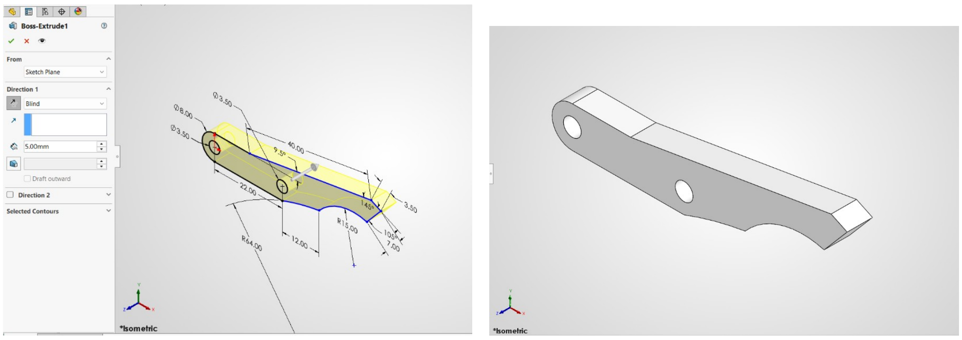

Figure 3.2: Snapshots while Designing Part 2 of Object 1 in Solidworks

Part 3

A cylindrical element with specific geometric features was modeled to facilitate rotational and translational motion within the assembly. Extrude and revolve features were used to create the core geometry, while bore holes were incorporated using pocket features. The part was designed with appropriate surfaces for bearing interfaces and load distribution.

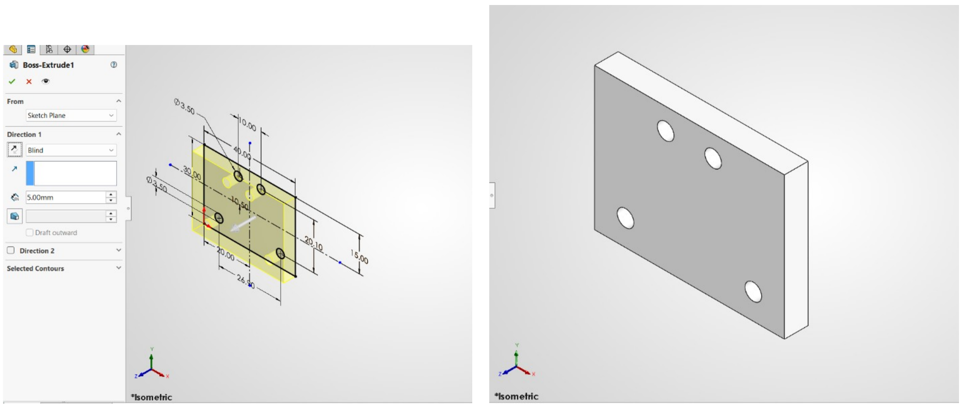

Figure 3.3: Snapshots while Designing Part 3 of Object 1 in Solidworks

Part 4

This auxiliary component was created to provide structural support and alignment for the assembly mechanism. Sketch-based features and simple extrusions were used to establish its geometry, with dimensional constraints ensuring compatibility with the other parts. Surface finish specifications and mounting points were carefully defined.

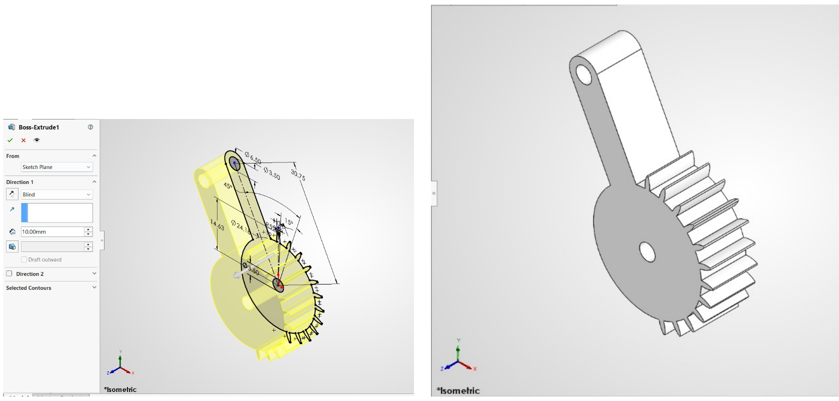

Figure 3.4: Snapshots while Designing Part 4 of Object 1 in Solidworks

Assembly

All four components were assembled using coincident and distance constraints to define the relative positions and allow for controlled motion between parts. Kinematic constraints were verified to ensure that the assembly mechanism functions as intended without interference or overlap. The final assembly was visualized from multiple angles to confirm proper alignment and motion characteristics.

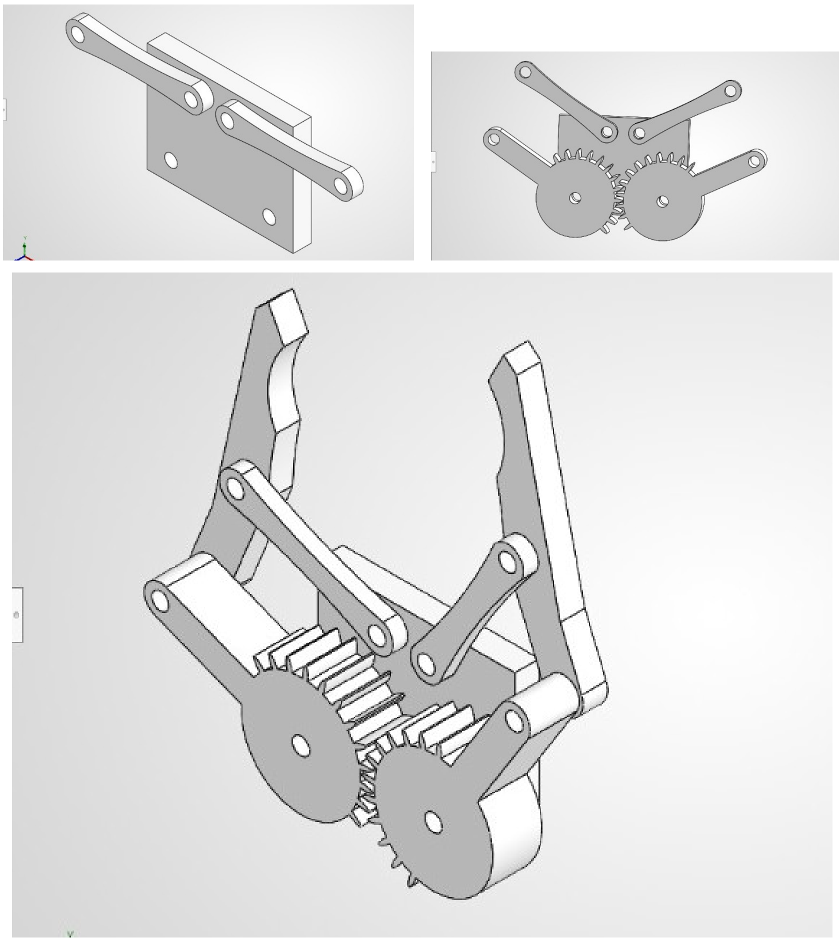

Figure 3.5: Snapshots while Assembling Object 1 in Solidworks

3.5 Object 2

Part 1

The first component was designed with a specialized profile that generates motion through geometric interaction with mating elements. Complex curved surfaces were created using sketch and extrude features with precise dimensional control. The part geometry was carefully refined to ensure smooth operation and efficient energy transmission throughout the operational cycle.

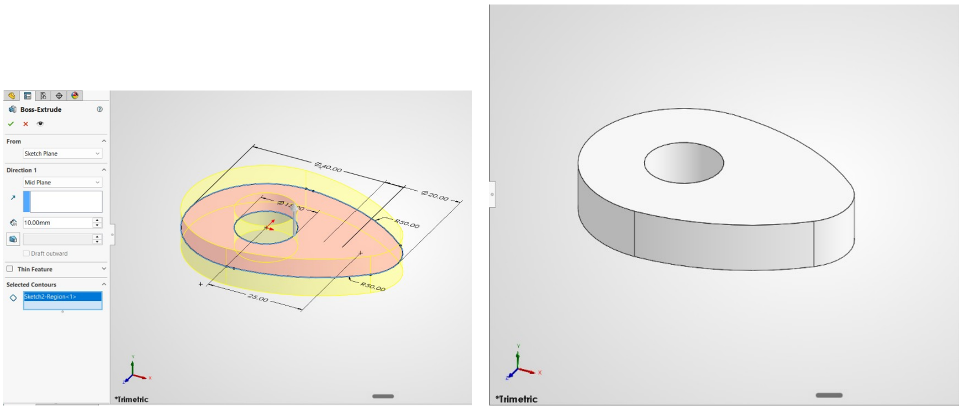

Figure 3.6: Snapshots while Designing Part 1 of Object 2 in Solidworks

Part 2

This component was modeled to respond to the geometric inputs provided by the primary part, converting its motion into linear or rotational output. A cylindrical body with appropriate contact surfaces was created using revolve and bore features. The design included specific dimensions and surface characteristics to ensure smooth interaction and minimize wear during operation.

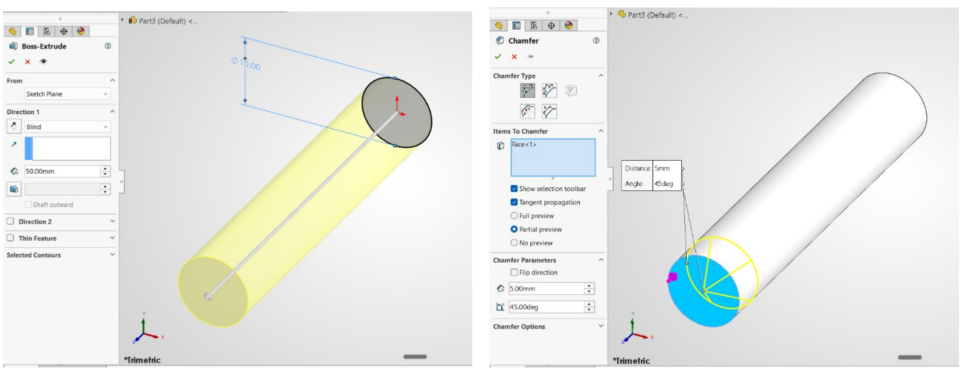

Figure 3.7: Snapshots while Designing Part 2 of Object 2 in Solidworks

Part 3

A structural support element was designed to constrain the motion of the primary components and maintain proper spatial relationships. Extrusion and pocket features were employed to create mounting surfaces and guide slots for other parts. Appropriate clearances were incorporated to allow smooth motion while maintaining rigidity of the overall structure.

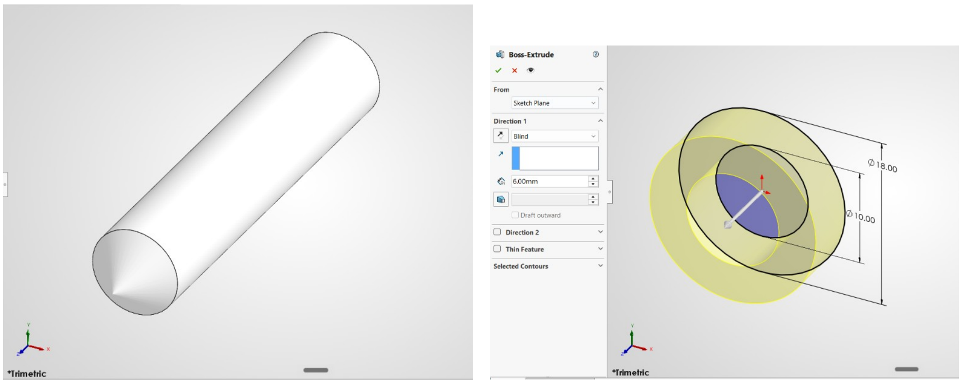

Figure 3.8: Snapshots while Designing Part 3 of Object 2 in Solidworks

Assembly

The three components were assembled with coincident and parallel constraints to establish the kinematic linkages required for proper mechanism function. The assembly was analyzed to verify that the motion sequences are correct and no component interference occurs during operation. Motion simulation confirmed that the mechanism achieves the intended output motion as designed.

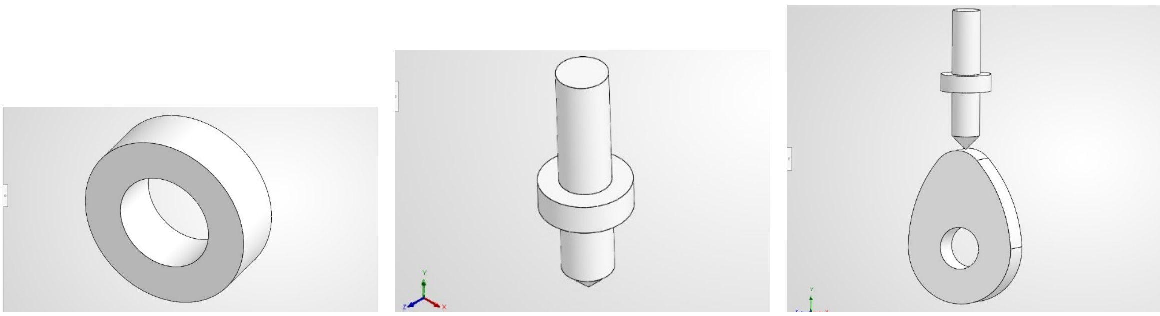

Figure 3.9: Snapshots while Assembling Object 2 in Solidworks

3.6 Discussion

The design and modeling of mechanical assemblies in SolidWorks provided valuable practical experience in translating conceptual designs into detailed parametric models. Both assemblies demonstrated the importance of precise dimensional control, proper constraint application, and systematic part modeling strategies. The parametric nature of the CAD system allowed for efficient modifications and ensured consistency across all components, enabling rapid design iteration and validation. Careful attention to detail during the sketching and feature creation phases proved critical to achieving functional assemblies that operate smoothly without component interference. The assembly constraints effectively simulated the kinematic behavior of the mechanisms, confirming that the geometric relationships were correctly defined. The experience highlighted how modern CAD tools enable engineers to verify designs virtually before physical fabrication, reducing costly errors and iterations in the manufacturing process.

3.7 Conclusion

The successful completion of this lab demonstrates proficiency in using SolidWorks for mechanical component design and assembly creation. Both assemblies were modeled with appropriate geometric features, dimensional constraints, and assembly relationships, resulting in functionally complete mechanisms ready for technical documentation. The parametric design approach enabled efficient part development and modification, while the assembly constraints verified kinematic correctness. Through this exercise, essential skills in feature-based modeling, constraint application, and assembly design have been developed and reinforced. The ability to create accurate digital representations of mechanical systems is fundamental to modern engineering practice, and this lab provided practical experience in these critical competencies. The technical documentation generated, including detailed part drawings and assembly specifications, demonstrates the complete workflow from conceptual design through to manufacturing-ready designs.

References

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